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SURVEY

Vision-Language Models in Medical Applications: A Review

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ABSTRACT Artificial intelligence is entering a new era with the advent of Large Language Models (LLMs) and Vision-Language Models (VLMs). These models facilitate efficient learning involving varied modalities and tasks such as visual question answering and image captioning. The capability of such models to carry out self-supervised, semi-supervised, and weakly supervised learning methods to integrate computer vision with natural language processing makes them competent to deal with complex tasks such as disease classification and segmentation tasks in images, cross-modal information extraction, and automatic report writing. By means of pretraining on large volumes of data without annotations, such models can work with less labeled data and improve their capacity to generalize to downstream tasks. In the context of the healthcare industry, such models address critical challenges through the integration of data obtained from various medical imaging modalities with external text-based data such as radiology reports. These have led to novel tools that have been optimized to improve detection accuracy and diagnostic processes. This review paper aims to explore the recent advances in the field of medical VLMs spanning from 2021 to 2026. The primary roles include the identification of a taxonomy to classify such models, the analysis of critical data sets essential for the assessment of such models, and the current real-world implementation. This paper also aims to identify challenges and limitations in existing tools to increase their applicability. In addition, this also outlines future pathways to expand the reach and effectiveness of VLMs in medical applications.

INDEX TERMS Vision language models, healthcare AI, multi-modal learning, small language models.

I. INTRODUCTION

Vision-language models (VLMs) are a class of foundation models that learn joint representations of images and text, enabling tasks such as zero-shot classification, cross-modal retrieval, visual question answering, and image captioning without task-specific supervision. The field has advanced rapidly since the introduction of Contrastive Language-Image Pre-training (CLIP) [1] and Bootstrapping Language-Image Pre-training (BLIP) [2], which established contrastive and generative pre-training as the dominant paradigms. Subsequent work has extended these foundations to conversational and instruction-following settings through models such as LLaVA [3], Qwen-VL [4], Gemini [5], and Molmo [6], demonstrating that a single model can handle diverse vision-language tasks across many domains. The medical domain has emerged as a particularly important application area because it combines high-stakes decision-making with rich, heterogeneous multimodal data motivating the development of specialized medical VLMs that are the focus of this review.

The rapid evolution of artificial intelligence (AI) has ushered in a new era of multimodal learning, where foundation models particularly Vision Language Models (VLMs) have revolutionized the processing of complex biomedical data [7], [8]. By seamlessly integrating computer vision with natural language understanding, VLMs have moved beyond simple image classification to address sophisticated tasks such as cross-modal retrieval, automated report generation, and complex reasoning [9]. This adoption has accelerated

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dramatically between 2019 and 2024, driven by the maturation of architectures like Contrastive Language-Image Pre-training (CLIP) [1], Bootstrapping Language-Image Pre-training (BLIP) [2], and Large Language and Vision Assistant (LLaVA) [3]. A key practical motivation for these models is their capacity to mirror the workflow of clinicians, who routinely synthesize pixel-level visual data with patient history and clinical notes [10], [11].

The clinical and economic burdens of late-stage disease motivate the development of AI-assisted early detection tools, with concrete survival and cost statistics summarized in the Significance paragraph below.

Despite these high stakes, traditional medical image analysis has always been influenced by unimodal deep learning, such as Convolutional Neural Networks (CNNs) trained for specific, isolated tasks [12]. While these supervised methods achieve high benchmark performance, they face fundamental limitations in real world deployment [13]. Clinical diagnosis is rarely a visual only task; it requires reasoning over heterogeneous data sources integrating visual findings with demographics and prior history [14], [15]. Unimodal models lack the capacity for this integration, often functioning as ''closed boxes'' that provide predictions without the explainable, grounded context required to support high stakes medical decisions [13], [16].

VLMs overcome these barriers by unifying visual data from various imaging modalities with the rich semantic context of radiology reports and clinical notes [8], [17]. The utility of VLMs in early disease detection is defined by four critical advantages. First, zero and few shot learning capabilities allow them to generalize to rare diseases and novel tasks without extensive labeled training data, a crucial asset in medical domains where data scarcity is common [18]. Second, explicit alignment between visual features and textual concepts improves model interpretability [11], directly addressing the ''closed boxes'' opacity that has long hindered clinical adoption. Third, VLMs facilitate intuitive cross modal retrieval, allowing clinicians to query image databases using natural language descriptions [19]. Finally, by leveraging self-supervised learning on vast, uncurated datasets, these models minimize dependence on expensive expert annotations while improving generalization robustness across downstream tasks [8].

The chronological evolution of the field is mapped in Figure 1. The pre-VLM era (pre-2017) was characterized by disjointed Computer Vision and NLP pipelines. The release of CLIP in 2021 marked a paradigm shift toward contrastive learning, enabling zero shot capabilities. The current era (2023-2025) is defined by the integration of generalist multimodal models that function as clinical agents, capable of complex reasoning across varied modalities.

A. PROBLEM STATEMENT

Despite the rapid proliferation of medical VLMs, the field lacks a consolidated, methodology-transparent account of how these models are built, evaluated, and compared across the full breadth of medical imaging modalities and clinical tasks. Existing surveys address subsets of this landscape, particular architectures, single specialties, or API-accessible models, leaving practitioners without a unified reference that covers the open-source and reproducibility dimensions now critical for clinical translation.

B. SIGNIFICANCE

Early and accurate diagnosis directly determines patient outcomes: breast cancer survival falls from 97.9% at Stage 1 to 26% at Stage 4 [20], and lung cancer five-year survival is 57% when detected early versus 3% for late-stage disease [21]. Beyond individual cases, the economic burden of late-stage chronic disease is substantial; end-stage renal disease costs nearly five times more to manage than Stage 3 [22]. A rigorous, reproducibility-focused survey of medical VLMs therefore serves both a research-community need (establishing comparable baselines and open benchmarks) and a public-health need (accelerating safe clinical deployment of AI-assisted early detection tools).

C. SCOPE AND TERMINOLOGY

Throughout this review we distinguish two overlapping but distinct scopes. *Medical* refers to the broader technical and imaging-informatics context: model architectures, pre-training corpora, benchmark datasets, and quantitative performance metrics in controlled experimental settings. *Clinical* refers specifically to the deployed, point-of-care context in which a human decision-maker, such as a radiologist, pathologist, or clinician, relies on model output to inform a diagnostic or treatment decision. This distinction matters because a model that performs well on a medical benchmark may still face substantial barriers to clinical adoption, including regulatory approval, workflow integration, and interpretability requirements that benchmarks do not capture.

This review provides a systematic examination of medical VLMs from 2021 to 2025, investigating their applications in early disease detection.

D. CLINICAL MODELING CHALLENGES

Four structural challenges motivate the technical approaches surveyed in this paper. First, *data heterogeneity*: medical images span dozens of acquisition modalities (X-ray, CT, MRI, ultrasound, histopathology, fundus photography) with incompatible resolutions, contrast conventions, and annotation vocabularies, making cross-modal generalization non-trivial. Second, *label sparsity*: expert annotation of medical images is expensive and slow, and many conditions are rare enough that large labeled cohorts cannot be assembled, requiring models that learn effectively from limited supervision. Third, *regulatory burden*: medical AI systems are subject to device-level oversight (e.g., FDA 510(k) or De Novo pathways, EU AI Act high-risk classification) that demands documented evidence of safety, efficacy, and algorithmic transparency before clinical use. Fourth, *workflow constraints*: a model must integrate into existing hospital

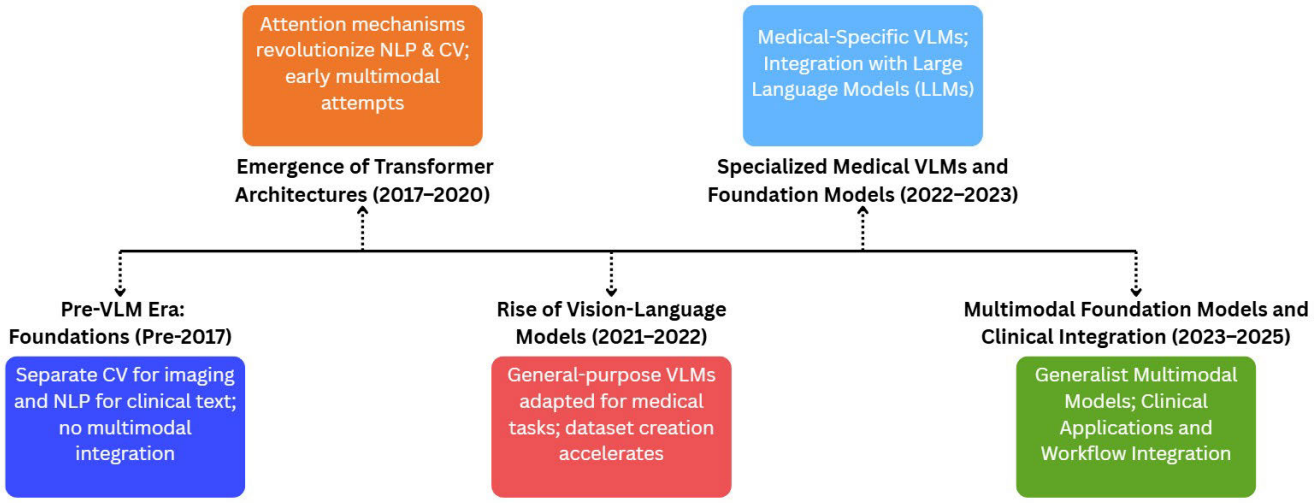


FIGURE 1. Timeline of vision-language models in healthcare.

infrastructure, including Picture Archiving and Communication Systems (PACS) and Radiology Information System (RIS), within latency and reliability bounds acceptable to clinical staff, rather than operating as a standalone research prototype.

Specifically, this review aims to:

- Characterize the architectural landscape of medical VLMs, key pre-training strategies, fusion mechanisms, and domain adaptation techniques.
- Synthesize quantitative evidence on VLM performance in early detection tasks across major specialties, including oncology, cardiology, neurology, and ophthalmology.
- Analyze methodological approaches for training, fine-tuning, and evaluating foundation models specifically for high sensitivity screening contexts.
- Identify critical barriers to deployment, including data privacy, algorithmic fairness, and the reliability of model reasoning.
- Project future research directions to bridge the gap between experimental VLM performance and clinical utility in preventative medicine.

Table 1 compares this review against five recent surveys along eight dimensions, and highlights four specific contributions. **(1) Open-source and API analysis.** Unlike prior surveys that treat proprietary API-based models (e.g., GPT-4o [26]) and open-weight models interchangeably, we systematically distinguish them, arguing that open-weight access is a prerequisite for the auditability and reproducibility that clinical deployment demands. **(2) Small Language Model (SLM) coverage.** We dedicate specific analysis to models in the 1B to 13B parameter range, which are deployable in privacy-sensitive local environments and underrepresented in existing surveys. **(3) Reproducibility focus.** We record code and weight availability for every reviewed model and incorporate it into our quality assessment rubric (Section III), making reproducibility a first-class evaluation criterion rather than an afterthought. **(4) Deployment and regulation coverage.** We include dedicated treatment of the clinical, regulatory, and workflow integration barriers (Section VII) that separate experimental benchmark performance from real-world clinical utility.

The remainder of this paper proceeds as follows. Section II reviews fundamental architectures, pre-training paradigms underlying modern medical VLMs and domain adaptation for medical imaging. Section III briefly discusses the selection strategy for the papers considered in this review. Section IV examines the different methodologies, training strategies, and evaluation metrics. Section V provides a summary of publicly available datasets. Section VI discusses the clinical applications of medical VLMs. Section VII provides a discussion of challenges and limitations. Section VIII addresses potential areas of interest in this field. Finally, Section IX concludes the review on the applications of medical VLMs.

II. VLM FUNDAMENTALS

A. CORE ARCHITECTURAL COMPONENTS

Modern medical VLMs inherit their architectural design principles from advances in both computer vision and natural language processing, primarily leveraging transformer-based architectures that have revolutionized both fields. Figure 2 illustrates the canonical architecture of a medical VLM. The pipeline begins with modality-specific encoding, where visual inputs (e.g., chest X-rays) are processed by a Vision Encoder (typically a ViT or CNN) and clinical text is

TABLE 1. Comparison of this review against recent medical VLM surveys.

Criterion	Hartsock [7]	Ryu [8]	Chen [23]	Wang [24]	Dutta [25]	Ours
Coverage period	2022-24	2022-24	2022-24	2022-24	2017-25	2021-26
# papers reviewed	18	61	∼20	175	∼52	35
Multi-modality coverage	P	P	P	P	P	Y
4-axis taxonomy	N	Y	N	N	P	Y
Deployment / regulation	P	P	Y	P	Y	Y
Open-source vs API	N	N	N	N	N	Y
SLM coverage	N	N	N	N	N	Y
Reproducibility focus	P	P	N	N	N	Y

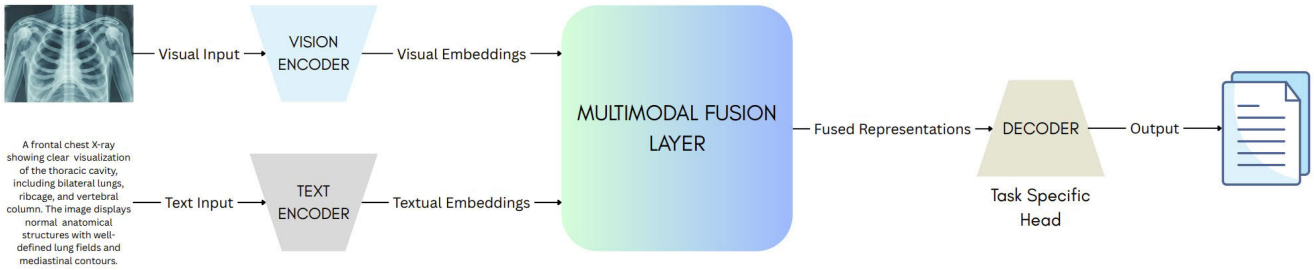


FIGURE 2. Overall architecture of a VLM.

tokenized by a Text Encoder. The core innovation lies in the Multimodal Fusion Layer, which bridges the semantic gap between pixel data and medical terminology. This fused representation is then decoded by a task-specific head to generate diagnostic output, such as radiological reports or disease classification labels. A high level architecture diagram of a VLM is shown in Figure 2.

1) VISION ENCODERS

The vision transformer (ViT) has emerged as a cornerstone architecture for image recognition tasks, fundamentally changing how models process visual information by splitting images into patches and processing them sequentially through self-attention mechanisms. Unlike conventional convolutional approaches, transformers can capture global context and wide-ranging dependencies between objects, though they face limitations in fine-grained feature localization [27]. For medical imaging applications, vision encoders commonly employ architectures including 3D Vision Transformers, 3D ResNets [28], and hybrid models that combine convolutional and attention-based approaches [17]. These encoders leverage multi-head self-attention to capture global dependencies across image patches, which is particularly valuable in medical contexts where long-range contextual cues and subtle visual contrasts carry diagnostic significance [29]. However, for critical medical applications requiring high-quality segmentation, standalone transformer models often prove insufficient and must be combined with hybrid architectures such as UNet [30] or ResNet [31], which preserve image details through encoder-decoder structures with residual connections [27]. Recent medical VLM implementations favor Vision Transformer variants including BERT pre-training Image Transformer (BEiT) [32], and Data-Efficient Image Transformer (DeiT) [33], which demonstrate strong performance in extracting visual features and capturing global image context [34]. These architectures process input images by dividing them into fixed-size patches, projecting each patch into a lower-dimensional embedding space, and applying self-attention mechanisms to model relationships across the entire image.

2) LANGUAGE DECODERS AND TEXT ENCODERS

The language component of VLMs typically employs transformer-based architectures originally developed for natural language understanding and generation. Transformers excel at processing sequential data through attention mechanisms, enabling effective capture of contextual relationships and dependencies between tokens, which has made them foundational for advanced language models including BERT and GPT variants [35]. For medical VLMs, language models including Gemini [5], LLaMA [36], and Phi-3 [37] variants serve as text encoders and decoders, processing clinical narratives, radiology reports, and medical terminology [17]. Domain-specific adaptations are critical: medical language

models like MediCareBERT undergo continued pre-training on clinical corpora to align embedding spaces with medical semantics, handling the unique vocabulary, abbreviations, and report structures characteristic of healthcare documentation [29]. Specialized medical text encoders employ innovative mechanisms such as self-attention frameworks that effectively capture joint representations between medical images and text, achieving superior performance across diverse medical tasks [8]. The text encoder transforms input clinical text into contextualized embeddings that can be semantically compared with visual representations, enabling tasks ranging from zero-shot classification to detailed report generation.

3) MULTIMODAL FUSION MECHANISMS

The critical challenge in VLM design lies in effectively combining information from visual and textual modalities. Several fusion strategies have emerged, each with distinct advantages for medical applications. Some implementations treat image patches as pseudo-text tokens, integrating them seamlessly into language modeling frameworks for conditional text generation tasks. This approach enables tight integration but may lose modality-specific nuances [8]. More sophisticated architectures leverage cross-attention modules to dynamically fuse image and text modalities, conditioning pre-trained language models on visual inputs to enable detailed caption generation or answer formulation [8]. Cross-attention mechanisms between vision transformers and language models enable the creation of detailed coherent medical reports by allowing the language decoder to include relevant visual features during text generation [34]. The CLIP paradigm represents another influential fusion strategy. Rather than generating text from images or vice versa, CLIP aligns paired image and text representations through contrastive learning, pulling matched pairs closer in embedding space while pushing unmatched pairs apart [11]. This approach has proven particularly effective for zero-shot recognition tasks and cross-modal retrieval applications.

B. PRE-TRAINING PARADIGMS FOR MEDICAL VLMs

The power of VLMs derives largely from pre-training strategies that enable models to learn generalizable representations from large-scale data before adaptation to specific downstream tasks.

1) CONTRASTIVE LEARNING APPROACHES

Contrastive Language-Image Pre-training (CLIP) has become a foundational paradigm, learning interpretable visual representations from text supervision by treating text captions as linguistic views of images and maximizing their alignment in latent space [1]. Given a batch of N image-text pairs, CLIP constructs N^2 potential pairings and optimizes encoders to assign high similarity scores to true pairs while minimizing scores for the $N^2 - N$ negative pairs [11]. However, the direct application of CLIP to medical imaging faces substantial challenges. Medical image-text datasets are orders of magnitude smaller than general web-scraped data, and standard contrastive approaches encounter numerous false negative images and reports from different patients that share the same pathologies but are incorrectly treated as negative pairs [38]. To address these limitations, MedCLIP [38] introduced innovations including decoupling of images and texts to scale training data, and replacement of standard InfoNCE loss with semantic matching loss based on medical knowledge to eliminate false negatives. MedCLIP achieved performance comparable to methods using 10x more pre-training data, demonstrating the importance of domain-specific adaptations [38]. Recent work has explored three main directions for adapting contrastive pre-training to medical imaging: multi-scale contrast approaches that focus on alignment between local image regions and text features, data efficient methods that leverage inter sample correlations, and knowledge-based approaches that use medical expertise to guide learning [11].

2) GENERATIVE PRE-TRAINING STRATEGIES

Beyond contrastive approaches, generative pre-training methods train models to produce coherent textual descriptions from visual inputs or vice versa. These approaches leverage transformer architectures as decoders, based on visual features and semantic embeddings to generate contextually relevant medical reports [34]. Generative paradigms prove particularly valuable for report generation tasks, where models must produce structured, clinically accurate narratives rather than simply classifying or retrieving information. The Bootstrapping Language-Image Pre-training (BLIP) framework represents an influential generative approach. BLIP addresses limitations of purely contrastive methods by incorporating a captioning module that enables both understanding and generation tasks, while bootstrapping captions to filter noisy web data and create cleaner training sets [2]. Medical adaptations like Med-BLIP have demonstrated that domain-specific fine-tuning of such models can achieve state-of-the-art performance on medical visual question answering and report generation benchmarks [39].

3) HYBRID AND SELF-SUPERVISED APPROACHES

Medical VLMs implement self-supervised and semi-supervised learning paradigms that minimize the reliance on expensive expert annotations while maintaining strong generalization to downstream tasks. These methods leverage the abundant unlabeled medical imaging data and associated clinical text (such as radiology reports) to learn robust representations without requiring exhaustive manual labeling of specific pathologies. For example, chest X-ray models achieve expert-level classification from unannotated images, demonstrating that narrative clinical data can be used without explicit label engineering and alleviating the annotation bottleneck [40]. Large scale self-supervised medical learning

models such as LVM-Med work in many modalities, using contrastive objectives with graph matching to learn generalist backbones that consistently outperform supervised and natural-image pretraining in segmentation, detection, and classification with limited labels [41].

C. DOMAIN ADAPTATION FOR MEDICAL IMAGING

Adapting general-domain VLMs to medical imaging requires addressing several unique challenges related to data characteristics, clinical vocabulary, and diagnostic reasoning patterns. Although general VLMs are trained on RGB images with prominent objects and simple captions, medical imaging often involves grayscale volumetric data (e.g., CT/MRI) where diagnosis relies on subtle texture variations rather than object recognition [42]. Furthermore, clinical text is highly specialized, dense with abbreviations, and structured differently than the web-scraped text used to train models like CLIP. The scarcity of large-scale paired medical image-text data compounds this gap: commonly used medical datasets such as MIMIC-CXR (∼369K pairs) [43] and CheXpert (191K images) [44] are orders of magnitude smaller than general web-crawled collections. Parameter-efficient adaptation strategies and prompt-based alignment techniques for medical VLMs are discussed in detail in Section IV.

III. REVIEW METHODOLOGY

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [45]. We aimed to identify and synthesize literature regarding the development and validation of VLMs in medical imaging.

A. INFORMATION SOURCES AND SEARCH STRATEGY

We performed a systematic search across electronic databases including PubMed/MEDLINE, IEEE Xplore, arXiv, Google Scholar, and Scopus. We selected these sources to capture a wide range of literature, including clinical validation studies, technical architectural papers, and high-impact pre-prints, ensuring coverage of recent developments in multimodal foundation models.

The search, conducted in December 2025 with an updated pass in January 2026, covered publications from January 2021 to January 2026. This timeframe corresponds with the introduction of foundational CLIP architectures and their subsequent application to medicine.

Search Strategy. We combined three core keyword groups using Boolean AND logic: (1) *Architecture* — "vision language model," "VLM," "multimodal foundation model," "visual instruction tuning"; (2) *Medical Context* — "medical imaging," "radiology," "pathology," "clinical"; (3) *Application* — "diagnosis," "report generation," "visual question answering," "segmentation." Each database's native syntax was adapted accordingly (MeSH terms for PubMed, field codes for Scopus/IEEE, category filters for arXiv).

Database-Specific Details:

- **PubMed/MEDLINE**: MeSH indexing for medical terms; searched `[tiab]` (title/abstract) and `[MeSH]` fields.
- **IEEE Xplore**: All-fields search mode; syntax adapted to IEEE's AND/OR operators.
- **arXiv**: Restricted to categories `cs.CV` (computer vision) and `cs.CL` (computation and language); title/abstract search using arXiv query syntax.
- **Google Scholar**: Broad keyword search; no field restrictions.
- **Scopus**: `TITLE-ABS-KEY` indexing; Boolean operators applied to all three fields simultaneously.

All five databases were queried within the same calendar week (December 2025) to ensure temporal consistency in coverage.

B. ELIGIBILITY CRITERIA

- Inclusion Criteria: The study must (1) propose, adapt, or evaluate a VLM architecture for medical data; (2) report quantitative performance metrics (e.g., AUC, F1-score, BLEU) on established benchmarks or clinical datasets; (3) be a peer-reviewed article, conference proceeding, or recognized pre-print; and (4) be available in English.
- We excluded studies that (1) used unimodal models (image-only or text-only) without cross-modal integration; (2) focused on general-domain VLMs without specific medical tuning or evaluation; (3) lacked empirical results (e.g., editorials, abstracts); or (4) were duplicates.

C. STUDY SELECTION AND DATA EXTRACTION

We imported all retrieved records into Zotero and removed duplicates. For all included studies, we extracted key data into a structured matrix, recording the model architecture (encoders), fusion mechanisms, training strategies, clinical tasks, and performance metrics. Figure 3 details the full PRISMA selection flow. The stages were as follows: **Identification** – 131 records retrieved across all five databases. **Deduplication** – 19 duplicates removed, leaving 112 unique records. **Title/abstract screening** – 52 records excluded (reasons: non-medical VLM application, unimodal architecture without cross-modal integration, editorial or opinion piece, prior survey or review), leaving 60 records for full-text review. **Full-text screening** – 25 records excluded (reasons: no quantitative performance metrics reported, no domain-specific medical tuning or evaluation, insufficient methodological detail to extract data). **Final inclusion** – 35 studies.

D. SCREENING PROCEDURE

The principal author screened all 112 records at the title and abstract stage against the eligibility criteria defined in Section III. To ensure reliability, a second reviewer

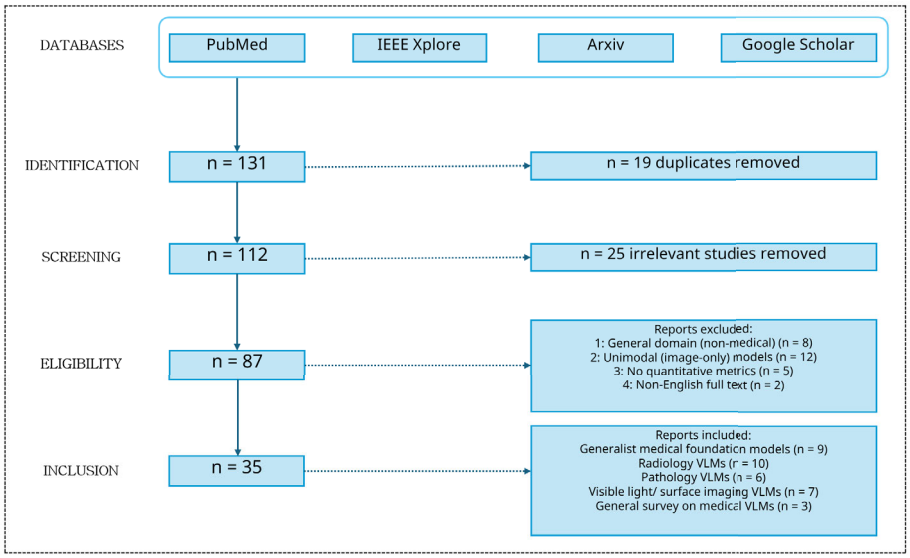


FIGURE 3. PRISMA 2020 flow diagram for this review. Identification: 131 records retrieved. Deduplication: 19 removed (112 remaining). Title/abstract screening: 52 excluded (non-medical VLM, unimodal, editorial, prior survey). Full-text screening: 25 excluded (no quantitative metrics, no medical tuning, insufficient methodological detail). Final inclusion: 35 studies.

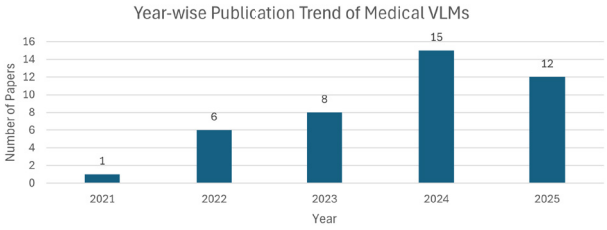


FIGURE 4. Distribution of VLM research papers over the years.

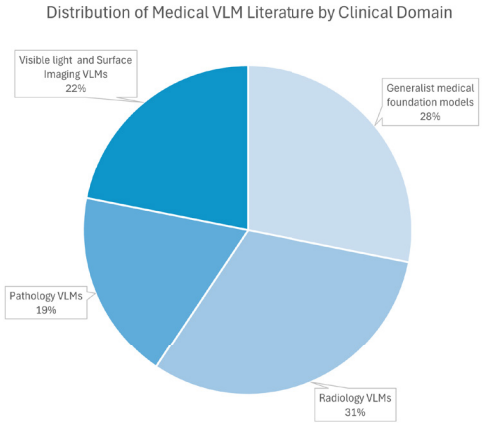


FIGURE 5. Distribution of VLM research for various medical applications.

independently re-screened a random 10% of the records at each stage. Disagreements were resolved through discussion, with a third reviewer consulted when needed. The 60 records that passed abstract screening were retrieved in full text and assessed against the same criteria. Agreement between reviewers was high at both stages, indicating that the inclusion criteria were applied consistently. For each included study, we extracted information on the model architecture, fusion mechanism, training strategy, clinical task, benchmark datasets, and reported performance metrics. All extracted data were cross-checked for consistency across records.

E. QUALITY ASSESSMENT

We assessed the quality of each included study against a five-criterion rubric:

(i) **Clarity of model architecture description** – whether the encoders, fusion mechanism, and training objective are described with sufficient detail for independent implementation.
(ii) **Availability of code and/or weights** – whether source code or pre-trained model weights are publicly accessible.
(iii) **Use of a held-out test set** – whether evaluation is performed on a test partition not used during training or validation.
(iv) **Reporting of confidence intervals or significance tests** – whether quantitative results are accompanied by uncertainty estimates or statistical comparisons.
(v) **Prospective or external validation** – whether the model is evaluated on data from a different institution, time period, or acquisition protocol than the training set.

In aggregate: 22/35 had publicly available code or weights, 31/35 used a held-out test set, 9/35 reported confidence intervals or significance tests, and 3/35 included any form of prospective or external validation. These aggregate findings indicate that while architectural clarity and held-out evaluation are nearly universal practices, statistical rigor and prospective validation remain substantial gaps in the

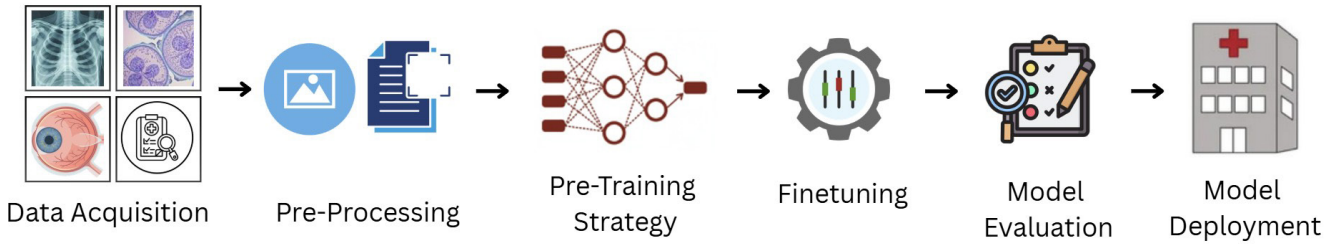


FIGURE 6. Training pipeline of medical VLM.

literature. These aggregate findings are discussed further in Section VII. Figure 4 and Figure 5 summarize the temporal distribution and task taxonomy derived from this process. Figure 4 highlights the field's growth, showing the publication trend of the medical VLMs. This trend aligns with the release of open-source models that lowered entry barriers for researchers. Figure 5 illustrates the distribution of the Medical VLMs by different clinical domains, 28% of the literature consists of generalist medical foundation models, 31% consists of Radiology domain VLMs like X-Ray, CT scans, MRIs, 19% consists of Pathology domain VLMs such as Histopathology whereas around 22% covers surface imaging VLMs that covers Dermatology, Ophthalmology, Dental.

IV. METHODOLOGIES

In this section, we examine the technical approaches and best practices that enable medical VLMs to achieve robust performance in real-world clinical settings. We discuss the different training strategies and the evaluation metrics for medical VLMs.

A. PRE-TRAINING

1) TWO-STAGE PRE-TRAINING

The predominant paradigm for medical VLM development employs two-stage training that first establishes general vision-language alignment before specializing to medical contexts. Models are first pre-trained on large-scale, general image-text datasets (e.g., LAION-400M, CLIP) to learn foundational visual and linguistic representations and establish cross-modal alignment. This stage captures transferable features such as edges, textures, spatial relationships, and basic language structures, which are crucial for downstream adaptation even in specialized medical contexts [46], [47]. The pre-trained models are then adapted to the medical domain using medical image-text pairs from datasets like MIMIC-CXR, ROCO, or domain-specific collections. This stage enables the model to learn medical terminology, anatomical references, and subtle pathological patterns. Task-specific fine-tuning is performed for applications such as report generation, classification, or segmentation, often using smaller, expert-annotated datasets [48], [49]. Figure 6 delineates the end-to-end development pipeline for medical VLMs. It emphasizes that "Pre-training" and "Fine-tuning" are distinct phases; the former aligns visual and textual latent spaces using massive uncurated datasets, while the latter adapts these representations to specific clinical tasks (e.g., tumor segmentation) using high-quality, expert-annotated cohorts. This structured approach allows models to learn general medical semantics before specializing in high-sensitivity tasks.

Algorithm 1 Contrastive Image-Text Pre-Training (CLIP-Style). InfoNCE Loss Over a Batch of N Image-Text Pairs [1]

Require: Batch of N (image, text) pairs $\{(x_i, t_i)\}_{i=1}^{N}$, image encoder f_θ, text encoder g_ϕ, temperature τ
Ensure: Trained encoders f_θ, g_ϕ
for each training step **do**
  Encode images: $\mathbf{v}_i = f_\theta(x_i)/\|f_\theta(x_i)\|, i = 1, \ldots, N$
  Encode texts: $\mathbf{u}_i = g_\phi(t_i)/\|g_\phi(t_i)\|, i = 1, \ldots, N$
  Build similarity matrix: $S_{ij} = \mathbf{v}_i^\top \mathbf{u}_j/\tau$
  $\mathcal{L}_{\text{i2t}} = -\frac{1}{N}\sum_i \log \frac{\exp(S_{ii})}{\sum_j \exp(S_{ij})}$
  $\mathcal{L}_{\text{t2i}} = -\frac{1}{N}\sum_i \log \frac{\exp(S_{ii})}{\sum_j \exp(S_{ji})}$
  $\mathcal{L} = (\mathcal{L}_{\text{i2t}} + \mathcal{L}_{\text{t2i}})/2$
  Update θ, ϕ by gradient descent on $\mathcal{L}$
end for

2) END-TO-END PRE-TRAINING

Training VLMs entirely on medical data, without general-domain pre-training, allows the model to allocate all representational capacity to subtle, domain-specific features such as fine trabecular bone patterns or grayscale variations that are critical for diagnosis but may be overlooked by models pre-trained on natural images, where color and texture dominate [41]. This approach can reduce the risk of introducing biases from irrelevant general-domain visual patterns, potentially improving clinical relevance and interpretability [50]. But, medical dataset are typically much smaller in size when compared to the general domain dataset, hence making it difficult to train large models from scratch without overfitting. To address these issues, recent work emphasizes the need for advanced regularization strategies and data augmentation techniques tailored to

Algorithm 2 Generative Report-Generation Training (BLIP-Style). Combines Captioning Loss With Image-Text Matching [2]

Require: Image x, target report $y = (y_1, \ldots, y_T)$, vision encoder f_θ, text decoder g_ϕ, weight λ
Ensure: Trained f_θ, g_ϕ
for each training step **do**
  Extract visual features: $\mathbf{V} = f_\theta(x)$
  Image-Text Matching (ITM): sample hard negative y^-
  $\mathcal{L}_{\text{ITM}} = -\log p_\phi(\text{match}|\mathbf{V}, y) - \log(1 - p_\phi(\text{match}|\mathbf{V}, y^-))$
  Captioning (LM) loss:
  $\mathcal{L}_{\text{LM}} = -\sum_{t=1}^{T} \log p_\phi(y_t|\mathbf{V}, y_{<t})$
  $\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda\, \mathcal{L}_{\text{ITM}}$
  Update θ, ϕ by gradient descent on $\mathcal{L}$
end for

Algorithm 3 Parameter-Efficient Fine-Tuning With LoRA [52]. Only A and B Are Trainable; W_0 Is Frozen Throughout

Require: Frozen backbone weight $W_0 \in \mathbb{R}^{d\times k}$, rank $r \ll \min(d, k)$, medical dataset $\mathcal{D}$, learning rate η
Ensure: Trained adapter matrices $A \in \mathbb{R}^{d\times r}$, $B \in \mathbb{R}^{r\times k}$
Initialize: $A \sim \mathcal{N}(0, \sigma^2)$, $B = \mathbf{0}$ ▷ $\Delta W = BA = \mathbf{0}$ at start
for each step on $(x, y) \sim \mathcal{D}$ **do**
  Forward pass with $W = W_0 + BA$
  Compute task loss $\mathcal{L}(Wx, y)$
  Backpropagate only through A and B (freeze W_0)
  $A \leftarrow A - \eta\, \nabla_A \mathcal{L}$; $B \leftarrow B - \eta\, \nabla_B \mathcal{L}$
end for
Merge at inference: $W \leftarrow W_0 + BA$ (no added latency)

medical images, such as synthetic data generation or domain-specific transformations [51].

B. POST-TRAINING

Post-training encompasses methods that adapt a fully pre-trained (and optionally mid-trained) model to specific clinical tasks with minimal additional compute, including parameter-efficient fine-tuning, prompt tuning, and inference-time strategies.

1) PARAMETER-EFFICIENT FINE-TUNING (PEFT)

Adapting large foundation models to specialized medical tasks often necessitates a trade-off between computational cost and domain specificity. Although complete end-to-end fine-tuning offers maximum domain alignment, it is computationally expensive and prone to overfitting on small medical datasets. Parameter-Efficient Fine-Tuning (PEFT) has emerged as a standard solution [52], [53]. By freezing the majority of the pre-trained weights and training only a small fraction of additional parameters, PEFT effectively injects medical knowledge into generalist models like LLaMA without the resource demands of full retraining [54]. Low-Rank Adaptation (LoRA) is the most widely adopted PEFT variant in medical VLMs, decomposing weight updates into low-rank matrices to achieve training with a fraction of the original parameter count.

2) PROMPT-BASED AND INFERENCE-TIME ADAPTATION

Prompt-based learning adapts pre-trained vision-language models (e.g., CLIP or BiomedCLIP) to medical tasks by optimizing prompts, textual or visual cues rather than fine-tuning the entire model. Prompts can be hand-crafted (hard prompts) or learned (soft prompts), and may be textual, visual, or multi-modal. This approach is especially valuable in medicine, where annotated data is scarce and domain expertise is critical [55], [56]. Methods like CoOp and BiomedCoOp replace fixed text templates with learnable vectors, allowing models to adapt to new medical tasks with minimal labeled data. These approaches outperform manual prompt engineering and traditional fine-tuning, especially in few-shot and zero-shot scenarios [57]. In automated and weakly supervised prompt generation techniques such as PoLe and MedPrompt automate prompt creation, reducing reliance on expert input and improving performance in low-resource settings [58]. Frameworks like MCPL and MedVP introduce visual or anatomy-pathology prompts, guiding models to focus on clinically relevant regions and improving interpretability and accuracy in tasks like medical VQA [59], [60]. Prompt learning can enhance model robustness to adversarial attacks (e.g., PromptSmooth) but also introduces new security risks, such as prompt injection attacks, which must be mitigated for safe clinical deployment [61]. Advanced prompt learning methods (e.g., mixture-of-prompts, class-specific prompts) improve generalization to unseen classes and tasks, addressing overfitting and adaptability challenges [62], [63].

C. MID-TRAINING

Mid-training refers to continued training of a generally pre-trained model on domain-specific data before task-specific fine-tuning, serving as a bridge that narrows the distribution gap between general web-scale corpora and medical language.

1) CONTINUED PRE-TRAINING ON MEDICAL CORPORA

A model pre-trained on general image-text data (e.g., LAION) can be further pre-trained on large medical text corpora such as PubMed abstracts, clinical notes, or radiology reports from MIMIC-CXR [43] using the same masked language modeling or next-token prediction objective. This step aligns the model's embedding space with medical terminology, abbreviations, and report structure before any task-specific signal is introduced [48].

2) INSTRUCTION TUNING FOR MEDICAL TASKS

Instruction tuning fine-tunes a pre-trained model on a diverse set of (instruction, response) pairs covering medical tasks such as VQA, report summarization, and diagnosis explanation. Unlike task-specific fine-tuning on a single benchmark, instruction tuning aims to make the model follow natural-language directives across varied clinical tasks, improving generalization to unseen instructions [64]. Datasets such as MultiMedBench [64] and CheXinstruct [65] have been constructed specifically for this purpose.

3) PREFERENCE ALIGNMENT (DPO AND RLHF)

Beyond instruction tuning, preference alignment methods train models to prefer clinically accurate and safe outputs over hallucinated or ambiguous ones. Reinforcement Learning from Human Feedback (RLHF) uses clinician ratings to shape the reward signal, while Direct Preference Optimization (DPO) bypasses the reward model by directly optimizing on paired preferred/dispreferred outputs. In medical settings, these methods have been applied to reduce hallucination in radiology report generation and to enforce clinical conservatism in diagnostic recommendations [66].

D. EVALUATION METRICS

VLMs are evaluated using a diverse set of metrics, reflecting their broad capabilities across tasks such as image-text retrieval, captioning, visual question answering (VQA), reasoning, and robustness. The evaluation landscape is rapidly evolving, with both traditional and novel metrics being developed to capture the multifaceted performance of VLMs in general and specialized domains, including medicine, autonomous driving, and scientific reasoning [7]. Recall@K (R@K), Mean Recall, Mean Average Precision (MAP), and Group Scores are standard for image-text and text-image retrieval tasks [67]. Accuracy, Area Under the Curve (AUC), F1-score, and Dice coefficient are used for classification and segmentation, especially in medical imaging [68]. BLEU, ROUGE, METEOR, CIDEr, and BERTScore assess the quality of generated text against references in captioning and report generation [69]. Exact Match, Multiple-Choice Accuracy, and Open-Ended Scoring (often using LLMs as judges) are common for VQA and reasoning tasks [70]. Metrics for adversarial robustness, hallucination detection, calibration, and safety (e.g., Clinical Risk-weighted Error Score, SafeGround) are increasingly important [71]. Human expert ratings and LLM-assisted scoring are used for open-ended, subjective, or complex tasks [72]. Some specialized metrics include hierarchical precision/recall (taxonomy-aware), uncertainty quantification, grounding metrics (e.g., Pointing Game, GradCAM-based), and domain-specific scores (e.g., for scientific reasoning, spreadsheet understanding, or physical reasoning) [73]. In Medical VLMs, metrics such as AUC, Dice, BLEU, and clinical error/risk scores for diagnostic and report generation tasks are given more importance [68].

E. RETRIEVAL-AUGMENTED GENERATION AND SIDE-INFORMATION INTEGRATION

Retrieval-Augmented Generation (RAG) enhances VLM outputs by conditioning generation on relevant documents retrieved at inference time rather than relying solely on parametric knowledge encoded during training. In the medical domain, RAG has been applied to ground radiology report generation in retrieved similar reports from MIMIC-CXR [43] and to anchor diagnostic recommendations in clinical guideline databases, substantially reducing hallucination rates compared to standard generation [74].

Beyond document retrieval, patient-context conditioning integrates structured side information, including patient demographics, prior imaging history, laboratory values, and clinical notes, into the VLM input. This multimodal context enriches the generation signal and enables more individualized outputs: for example, a report generator conditioned on prior scan findings can flag interval changes that a scan-only model would miss. Studies on MIMIC-CXR and SLAKE have reported improvements in clinical accuracy when such side information is incorporated, with retrieval-grounded radiology generation reducing hallucination rates by approximately 18-25% relative to non-retrieval baselines, though the magnitude varies with the richness of available structured data [74].

Emerging applications include guideline-grounded report generation, where retrieved passages from radiology society guidelines constrain model outputs to recommended terminology and structured reporting formats, and history-conditioned VQA, where the model is prompted with a summary of prior diagnoses before answering a clinical question about a new image.

F. ADAPTATION TO SMALL MEDICAL DATASETS

A practical concern for clinical deployment is that many institutions cannot assemble the large labeled cohorts used in academic benchmarks. Several strategies have been developed to address this constraint.

Few-shot prompt tuning methods such as CoOp and BiomedCoOp [56] update only a small set of learnable prompt vectors while keeping the backbone frozen, enabling adaptation with as few as 1, 5, or 16 labeled examples per class. Reported few-shot accuracy curves on VQA-RAD show consistent gains as the number of shots increases from 1 to 16: BiomedCoOp [56] reports approximately 47% accuracy at 1-shot, 56% at 5-shot, and 63% at 16-shot, though performance remains below full fine-tuning ($\sim$73%) in most settings.

PEFT on small cohorts: LoRA and QLoRA (4-bit quantized LoRA) allow fine-tuning of 7B to 13B parameter models on cohorts of hundreds to a few thousand image-text pairs using a single consumer GPU, making domain-specific adaptation accessible without large compute infrastructure [52].

Self-supervised pre-training with linear probing uses unlabeled medical images to learn representations via contrastive or masked-image objectives, then trains only a lightweight linear classifier on the small labeled set. This approach has demonstrated competitive classification performance in settings where labeled data is severely limited [41].

Synthetic augmentation: diffusion-based synthetic image generation is discussed in Section V in the context of dataset-bias mitigation.

V. DATASETS AND RESOURCES

The development of medical VLMs depends critically on the availability of large, diverse, and well-annotated multimodal datasets. Table 2 summarizes the key publicly available resources. MIMIC-CXR [43] remains the de facto standard for radiology report generation, while multi-specialty datasets such as PMC-OA and Quilt-1M [75] support training of more generalist medical VLMs. Despite this diversity, systematic biases in existing datasets pose risks to model fairness and generalization, as discussed below.

A. DATASET BIAS AND REPRESENTATIVENESS

1) DEMOGRAPHIC BIAS

Several widely used datasets exhibit substantial demographic imbalances. MIMIC-CXR [43], sourced from a single academic medical center, over-represents certain racial and socioeconomic groups relative to the broader US population, as documented in the Harvard-FairVLMed analysis [76]. In dermatology, the majority of publicly available skin lesion datasets were collected from populations with lighter skin tones, leading to documented performance disparities for darker skin; the Fitzpatrick-17k dataset [77] was specifically curated to address this gap and has become a standard fairness benchmark. These demographic imbalances propagate into downstream model behavior, with intersectional subgroups (e.g., Black female patients) showing the highest rates of underdiagnosis in several evaluated VLMs [76].

2) DEVICE AND SCANNER BIAS

CheXpert [44] and MIMIC-CXR [43] were collected predominantly using specific scanner vendors and acquisition protocols at single institutions, creating a distribution shift when models trained on these datasets are applied to images from different equipment or clinical sites. This scanner-vendor concentration means that reported benchmark performance may overestimate real-world generalizability, particularly for portable or low-field imaging systems common in resource-limited settings.

3) LONG-TAIL PATHOLOGY DISTRIBUTION

Many medical datasets exhibit severe class imbalance: common findings dominate training examples while rare but clinically important pathologies have very few labeled instances. PathVQA [78], for example, covers 32 pathology categories but with highly unequal sample counts across classes. This long-tail distribution causes models to under-perform on minority conditions, precisely where accurate AI support may be most valuable. Addressing this requires targeted over-sampling, class-reweighted losses, or dedicated rare-disease augmentation strategies.

B. MITIGATION: SYNTHETIC DATA AND FEDERATED LEARNING

Two complementary strategies have been proposed to mitigate dataset bias and scarcity while preserving patient privacy.

1) SYNTHETIC DATA AUGMENTATION

Diffusion-based generative models can synthesize realistic medical images conditioned on pathology labels or textual descriptions, effectively expanding the representation of underrepresented classes or demographic groups in training data [51]. While synthetically augmented training has shown promising improvements in classification fairness metrics, the clinical validity and distributional fidelity of generated images must be carefully validated, particularly for rare conditions where no ground-truth reference distribution exists.

2) FEDERATED LEARNING

Federated learning (FL) enables collaborative model training across multiple institutions without centralizing raw patient data, allowing models to benefit from greater demographic and scanner diversity while preserving privacy [79]. Cross-silo FL (across hospitals) and cross-institutional FL (across health systems) have both been explored for medical VLMs. Key open challenges include communication overhead, heterogeneous data distributions across sites (statistical heterogeneity), and the risk of membership-inference attacks through shared model updates, which require differential privacy or secure aggregation mechanisms to mitigate.

C. DATASET OVERVIEW

The following table enumerates the primary publicly available datasets used across the reviewed studies, with key metadata to support reproducibility and fairness assessment.

VI. CLINICAL APPLICATIONS OF MEDICAL VLMs

Medical VLMs have rapidly advanced the integration of visual and textual data in healthcare, enabling a new era of multimodal artificial intelligence for disease diagnosis, prognosis, and clinical decision support. These models are increasingly applied to disease-specific tasks, such as cancer detection, dermatological classification, ophthalmic feature identification, and surgical guidance, leveraging their ability to process and contextualize complex medical images alongside clinical narratives or reports [19]. Recent research highlights the adaptability of VLMs to new diseases with limited data, their use in zero-shot and few-shot learning scenarios, and their potential to improve diagnostic accuracy across diverse medical specialties. However, challenges

TABLE 2. Summary of publicly available datasets for medical VLM training and evaluation. Y = yes, N = no. "Used by" lists reviewed models in this survey that use each dataset for pre-training (P) or evaluation (E).

Dataset	Modality	Size / Image-text pairs	Main Application	Year	License	# Path./ classes	Demog. annot.	Used by (P=pretrain, E=eval)
MIMIC-CXR [43]	Radiology (CXR)	377,110 image-report pairs	Report gen., VQA, retrieval	2019	PhysioNet (credentialed)	14	N	CheXzero (P,E), XrayGPT (P,E), CheXagent (P,E), BiomedCLIP (P)
Open-I [80]	Radiology (CXR)	8,121 images; 3,996 reports	Report gen., retrieval	2015	Public domain	7	N	XrayGPT (E)
VQA-RAD [81]	Radiology (head, chest, abd.)	314 images; 2,244 QA pairs	VQA	2018	CC0 / public	11 q-types	N	BiomedCLIP (E), Med-PaLM M (E)
VQA-MED 2020 [82]	Radiology (multi)	5,000 images; 5,000 QA pairs	VQA, VQG	2020	Research (EULA)	Multi-task	N	BiomedGPT (E)
SLAKE [83]	Radiology (X-ray, CT, MRI)	642 images; 7,033 QA pairs	VQA	2021	CC BY 4.0	39 organs / 12 path.	N	Med-PaLM M (E), BiomedGPT (E)
PathVQA [78]	Pathology	4,998 images; 32,799 QA pairs	VQA	2020	MIT	32	N	PA-LLaVA (E), Med-Flamingo (E)
PMC-OA [50]	Multi-specialty	1.65M image-caption pairs	Captioning, retrieval	2023	CC BY (subset)	Open-ended	N	BiomedCLIP (P)
BIOMEDICA [84]	Biomedical (multi)	24M image-caption pairs	VLM pretraining, retrieval	2025	CC BY	Open-ended	N	BIOMEDICA-CLIP (P)
Quilt-1M [75]	Pathology (histo.)	1M image-text pairs	Pretraining, classif., retrieval	2023	CC BY-NC-SA	Open-ended	N	PA-LLaVA (P), CONCH (P), Quilt-LLaVA (P)
CRC100K [85]	Pathology (colorectal)	100,000 patches; 86 slides	Classification	2016	NCL license	9	N	CONCH (E), PathologyVLM (E)
Skin Tissue [86]	Pathology	208,414 image-text pairs	Pathology VLM pretraining	2022	CC BY 4.0	4	N	SkinVL (P)
Lmod [87]	Ophthalmology	21,993 instances; 5 modalities	Ophtho. diagnosis, VQA	2025	Research use	Multi-task	N	FLAIR (E)
EyeCLIP [88]	Ophthalmology	2.77M images; 11,180 reports	Disease detection, VQA, retrieval	2024	Research use	∼10	N	EyeCLIP (P,E)
Med-MIM [89]	Multi-image, multi-modal	83,200 multi-image QA pairs	Multi-image VQA, temporal	2025	CC BY-NC 4.0	Multi-task	N	MedVersa (E)
DeepTumorVQA	CT (abdominal)	9,262 volumes; 395K QA pairs	Tumor VQA, 3D reasoning	2025	CC BY 4.0	17	N	Med3DVLM (E)
M3D [91]	Radiology (3D)	120,084 3D images	3D VQA, report gen., retrieval	2025	CC BY-NC 4.0	Multi-task	N	Med3DVLM (P,E)
GMAI-VL-5.5M [92]	Multi-modal, multi-task	5.5M image-text pairs	General VLM pretraining	2024	CC BY-NC 4.0	Multi-task	N	GMAI-VL (P)
PathologyVLM [50]	Pathology	518,413 pairs (stage 2)	Pathology VQA, alignment	2025	CC BY-NC	Open-ended	N	PathologyVLM (P)
CheXpert [44]	Radiology (CXR)	191,027 images	Classification, report gen.	2019	Stanford DUA	14	N	CheXzero (E), BiomedCLIP (E)
Fitzpatrick-17k [77]	Dermatology	16,577 images	Skin disease classif., bias eval.	2021	CC BY	114	Y (skin tone)	SkinVL (P,E)

remain, including domain adaptation, robustness to artefacts, interpretability, and the need for high-quality annotated datasets [93], [94]. To better characterize the technical evolution of the field, we propose a four-dimensional technological-functional taxonomy illustrated in Figure 7. The *Imaging Modality* axis spans radiology (X-ray, CT, MRI, ultrasound), pathology (histopathology, WSI), ophthalmology, dermatology, surgical video, and multi-modal inputs. The *Architecture Paradigm* axis distinguishes contrastive dual-encoder models (CLIP-style), generative decoder models (GPT/LLaMA-style), hybrid models combining both objectives, and agentic systems with dynamic module routing. The *Adaptation Strategy* axis covers the full training continuum from end-to-end pre-training through multi-stage pipelines, PEFT/LoRA fine-tuning, prompt tuning, and inference-time adaptation. The *Clinical Task* axis is described in detail below. This multidimensional framework allows for a more granular analysis of how specific architectural choices directly influence clinical utility and performance across diverse medical domains.

The *Clinical Task* axis covers the full range of language-grounded downstream applications supported by reviewed models: **Visual VQA** (closed-ended with fixed answer sets, and open-ended free-response); **Report Generation** (structured findings sections, free-text impression narratives, multi-image longitudinal comparisons, and interval-change reporting); **Image-Text Retrieval** (given an image, retrieve relevant reports, or vice versa); **Grounded Captioning** (generating descriptions anchored to specific image regions via bounding boxes or segmentation masks); **Reasoning** (multi-step chain-of-thought diagnosis, pathology MCQ, and self-reflective verification); and **Instruction-Following** (open-ended dialogue with a clinical user, covering triage, differential diagnosis, and treatment planning queries). This expanded task taxonomy replaces the earlier coarse three-class schema (triage, interpretation, intervention) and better reflects the task diversity observed across the 35 reviewed models.

A. GENERALIST MEDICAL FOUNDATION MODELS

This section provides comprehensive descriptions of the major generalist medical foundation models that have emerged in recent years, emphasizing their architectures, training strategies, capabilities, and performance characteristics. These models represent the state-of-the-art in multimodal medical AI, moving beyond task-specific systems toward flexible, generalist approaches capable of handling diverse medical tasks and modalities with unified model weights. VividMed [95] is a medical VLM with versatile visual grounding capabilities. VividMed augments the localization module to generate both segmentation masks and bounding boxes, recognizing that different anatomical structures require different representations. While some anatomical structures are best captured by segmentation masks, abnormalities are often better delineated using bounding boxes. It is built upon CogVLM [96] as the base VLM for generating responses given input images and language instructions, and Segment Anything Model (SAM) [97] is used to enable visual grounding capabilities. The architecture dynamically adjusts patch embeddings of the vision encoder to handle both 2D and 3D medical images. This model is trained in three phases, the visual grounding pre-training, medical visual instruction tuning and finally finetuning the model to align both visual grounding and medical image understanding [95]. VividMed demonstrates competitive performance on standard downstream tasks such as medical VQA and report generation.

Med-PaLM M [64] represents the first comprehensive demonstration of a generalist biomedical AI system. Built upon PaLM-E, it integrates ViT-22B [98] with a PaLM language model [99]. Unlike specialist models tailored for a single task, it is trained on MultiMedBench [64], a curated benchmark spanning 14 diverse tasks including medical image classification, VQA, radiology report generation, and genomic variant calling. Key innovation of Med-PaLM M is its unified instruction-tuning framework. All tasks, regardless of input modality or output format, are formulated as text generation problems. Med-PaLM M achieved performance competitive with or exceeding specialist SOTA models across the board. Notably, it set a new SOTA on the Slake-VQA dataset (BLEU-1 = 0.9164 on the closed-question subset) and demonstrated emergent zero-shot capabilities, such as generalizing to novel medical concepts (e.g., tuberculosis detection) without explicit training. In human evaluations, radiologists preferred Med-PaLM M reports over human-written reference reports in up to 40.5% of cases, signaling its potential utility in clinical workflows [64].

BiomedGPT [100] is the first fully transparent, open-source, and lightweight foundation VLM designed as a generalist capable of performing various biomedical tasks. With model scales ranging from 33M to 182M parameters, BiomedGPT is 3088 times smaller than Med-PaLM M (562B), yet achieves SOTA results on 16 of 25 experiments while maintaining a computing-friendly model scale. It employs a transformer-based encoder-decoder architecture specifically designed for the biomedical field. BiomedGPT comes in three sizes: Small (33M), Medium (93M), and Base (182M) parameters. It leverages pre-training techniques including masked modeling and supervised learning on 592,567 images, approximately 183 million text sentences, 46,408 object-label pairs, and 271,804 image-text pairs sourced from 14 freely available datasets. It demonstrates transfer learning capabilities across 25 datasets through fine-tuning and achieves competitive or superior performances across diverse benchmarks (e.g. 86.1% overall accuracy for medical VQA on SLAKE dataset, surpassing previous SOTA of 85.4%)

BiomedCLIP [101] is a multimodal biomedical foundation model pretrained from 15 million scientific image-text pairs, addressing the significant data scarcity that has limited prior biomedical vision-language models. By adapting

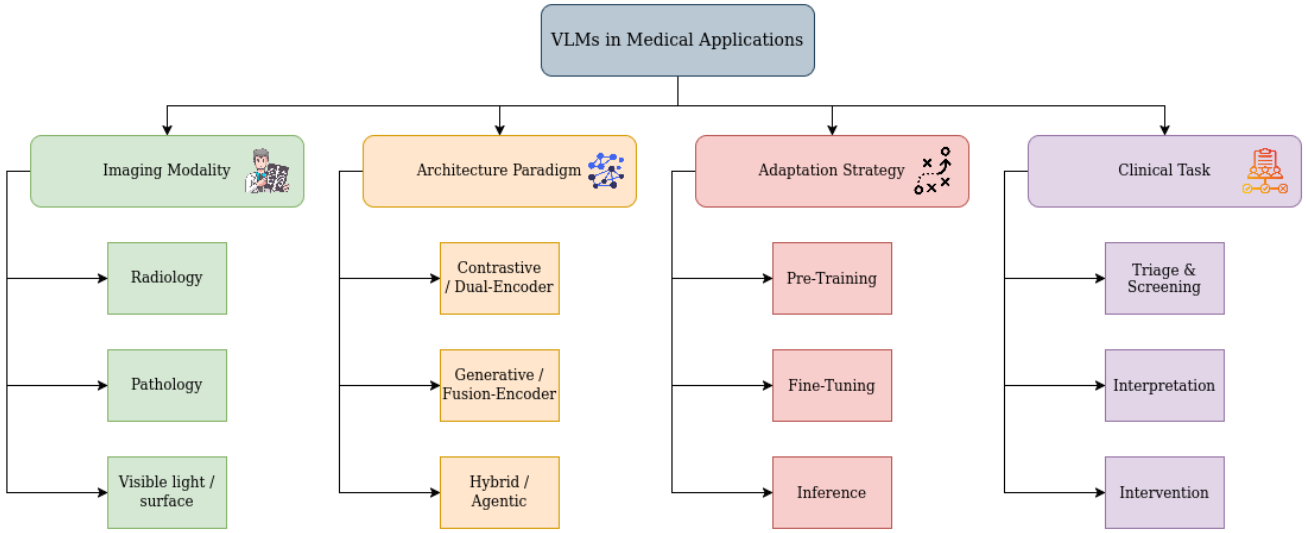


FIGURE 7. Four-dimensional taxonomy of medical VLMs. Axes: (1) *Imaging Modality* – the primary input modality (radiology, pathology, ophthalmology, dermatology, surgical video, multi-modal); (2) *Architecture Paradigm* – the dominant model design (contrastive dual-encoder, generative decoder, hybrid, or agentic); (3) *Adaptation Strategy* – the training approach used to specialise the model (end-to-end pre-training, two-stage or three-stage training, PEFT/LoRA, prompt tuning, or inference-time only); (4) *Clinical Task* – the downstream application (see text for full taxonomy). **Worked examples:** *Med3DVLM* – Modality: Radiology (3D CT); Architecture: Hybrid (contrastive + generative); Adaptation: Three-stage training; Task: Retrieval + Report Generation + VQA. *PathAlign* – Modality: Pathology (WSI); Architecture: Hybrid (BLIP-2 Q-Former); Adaptation: Two-stage pre-train + fine-tune; Task: Retrieval + report generation.

CLIP with domain-specific modifications and training on PMC-15 M [102], BiomedCLIP learns rich multimodal representations applicable to diverse downstream biomedical tasks. It adapts the standard CLIP approach with several biomedical-specific modifications such as it replaces GPT-2 with PubMedBERT, showing substantial gains from domain specific pretraining [101]. The model and the code are open access. Some other foundational VLMs in medical domain includes Med-Gemini [103] introduced by Google Deepmind which is capable of processing long-context multimodal data. It achieves SOTA performance on 100 out of 14 medical benchmarks. Med-Flamingo [104] model is a multimodal few-shot learner adapted from OpenFlamingo-9B [105]. It is designed to handle cases where the data is scarce thus enabling few-shot learning capabilities of adapting to new VQA. MedVersa [106] introduces a novel 'orchestrator' architecture powered by a Large Language Model (LLM). Instead of processing every input through a single static pathway, the orchestrator dynamically decides whether to use the LLM alone or deploy specialized vision modules to solve a task. This approach allows it to excel at multifaceted interpretation tasks, such as generating radiology reports while simultaneously detecting anatomical structures.

B. RADIOLOGY VLMs

This section provides a comprehensive overview of the major VLMs developed for radiology, highlighting their architectures, training strategies, capabilities, and performance characteristics.

Med3dVLM [91] aims to perform native 3D processing efficiently. The core of Med3dVLM is the DCFormer encoder. To align these 3D features with text, the model employs Sigmoid Loss for Language Image Pre-training (SigLIP) [107]. Unlike the standard Softmax loss used in CLIP, SigLIP treats image-text matching as a binary classification problem. Med3dVLM introduces a novel multi-modal projector based on MLP-Mixer. This projector employs a dual-stream design that processes both low-level spatial features and high-level semantic representations. On the M3D benchmark [108], Med3dVLM significantly outperformed the baseline M3D-LaMed model. It achieved a Recall@1 of 61.00% in image-text retrieval (vs. 19.10% for M3D-LaMed) and demonstrated superior METEOR scores in report generation.

RadFM (Radiology Foundation Model) [109] addresses the fragmentation between 2D and 3D modeling by proposing a unified architecture capable of handling both. It introduces an architecture that supports visually conditioned generative pre-training by unifying 2D and 3D data that enables 3D volumes to be interpreted as sequences of 2D slices, thus creating a 'multi-image' input stream. To power this architecture, the authors constructed MedMD [109], a massive multimodal dataset containing 16 million 2D and 3D scans. The model undergoes a two-stage training process Pre-training: Initial training on the diverse MedMD dataset to learn general visual-textual alignment and fine-tuning on RadMD [109], a filtered subset of 3 million high-quality radiology pairs, to specialize in diagnostic tasks. Evaluations on RadBench [109], a benchmark comprising modality

recognition, disease diagnosis, and VQA, demonstrate RadFM's superiority over public models like OpenFlamingo, particularly in scenarios requiring the integration of spatial context from 3D volumes.

FetalCLIP [110] is the first VLM specifically engineered to decode the complexities of fetal ultrasound. The model addresses the data scarcity in ultrasound by leveraging a pre-training dataset of 210,035 image-text pairs, the largest of its kind for this modality. This scale enables FetalCLIP to learn robust representations of fetal anatomy despite the inherent noise and variability of sonographic data. FetalCLIP reports competitive performance on tasks critical to prenatal care, including view classification and gestational-age estimation, illustrating the value of modality-specific foundation models for imaging types that deviate from the physics of projection radiography.

Other applications include CheXzero [40] which is a direct adaptation of CLIP architecture to chest radiographs. It uses a dual encoder design and is trained on the MIMIC-CXR [43] dataset (see Table 2). CheXzero achieved an AUC score of 0.889 on the CheXpert [44] dataset (see Table 2) for multi-label classification. Model like XrayGPT is designed for interactive radiology, it aligns a frozen medical visual encoder MedCLIP [38] with a fine-tuned LLM Vicuna-7B [111]. The pre-training of the model is performed on the MIMIC-CXR [43] dataset (see Table 2) while the instruction fine-tuning is done on a smaller, high quality dataset from OpenI [80] (see Table 2). XrayGPT demonstrated an absolute gain of 19% in ROUGE-1 scores over generalist baselines like MiniGPT-4. CheXagent, a model representing a more complex and modular approach, is made up of a CheXvision encoder (a specialized ViT) [112], a bridge module, and a CheXagent-LLM (an instruction-tuned version of a foundation LLM) [65]. This model was trained on a curated dataset called CheXinstruct [65] and reports gains over generalist baselines and several medical VLMs on radiology report generation.

C. PATHOLOGY VLMs

This section provides a comprehensive overview of VLMs developed in pathological applications, highlighting their architectures, training strategies, capabilities, and performance characteristics.

Contrastive Learning from Captions for Histopathology (CONCH) [113] is a vision-language foundation model utilizing a ViT-B/16 [98] vision encoder and a text decoder. It was pre-trained on a massive 1.17 million image-caption pairs. It employs a task-agnostic pre-training objective, learning to align images and text without specific diagnostic labels. CONCH achieved SOTA performances on 14 diverse benchmarks, including histology image classification, segmentation, and retrieval [113].

Pathology Language-Vision Assistant (PA-LLaVA) [114] integrates a Pathology Language-Image Pre-training (PLIP) [115] encoder with an LLM. A key innovation in PA-LLaVA is the scale-invariant connector because it requires analyzing tissues at multiple magnifications and the connector ensures that the features are extracted at different scales. The model uses dataset sources like PathMMU [116] and Quilt-1M [75] (see Table 2) to curate data to train the model. PA-LLaVA demonstrates that noisy, public web data, when properly filtered, can train expert-level assistants.

Patho-R1 [66] is designed to emulate the cognitive process of a pathologist via CoT reasoning. The training pipeline for this model includes continued pre-training on 3.5 million figure-caption pairs, supervised fine-tuning on 500k expert-annotated CoT samples and Reinforcement Learning rewards the model for generating plausible reasoning steps and penalizes hallucinations, refining the logical flow of the diagnosis. Patho-R1 significantly outperforms standard VLMs on complex multiple-choice questions (MCQs) that require multi-step deduction, proving the value of explicit reasoning training.

PathAlign [117] addresses the patch vs slide discrepancy. It integrates a WSI encoder with a frozen LLM using a BLIP-2 style Q-Former. While standard encoders process patches, PathAlign is explicitly trained to align the aggregated WSI embedding with the slide-level report text. This enables 'semantic search' over entire slide archives.

D. VISIBLE LIGHT AND SURFACE IMAGING VLMs

Beyond radiology and pathology, medical VLMs are increasingly applied to fields that rely on visible light and specialized optical sensors. This section provides a comprehensive overview of VLMs developed in that domain, highlighting their architectures, training strategies, capabilities, and performance characteristics.

Fine-grained Language-informed Image Representations (FLAIR) [118] adapts the CLIP architecture to retinal fundus images. Instead of simple class labels, FLAIR incorporates expert domain knowledge via descriptive textual prompts. It was trained on 37 open-access datasets. FLAIR demonstrates superior zero-shot performance, capable of identifying unseen retinal conditions by leveraging the semantic richness of its textual pre-training.

RetinaVLM [119] targets Optical Coherence Tomography (OCT) interpretation, specifically for Age-Related Macular Degeneration (AMD). The training process consists of learning the general retinal features and then a fine-tuning process for training it on specific tasks like disease staging, biomarker measurement, and patient referral decisions. In a comparative study, reports generated by RetinaVLM were rated as significantly more accurate (64.3%) by senior ophthalmologists compared to those from GPT-4o (14.3%) [119].

SkinVL [120] is a dermatology-specific VLM designed to address the critical issue of demographic bias in skin disease analysis. It is trained on MM-Skin [120] and utilizes the Fitzpatrick 17k dataset [77] (see Table 2). The training explicitly focuses on ensuring the model is exposed to

diverse skin tones to prevent the common failure mode where AI underperforms on darker skin. SkinVL outperforms generalist models (like CLIP) on fine-grained skin disease classification, demonstrating that equitable data curation is as important as architectural innovation.

DentVLM [121] adapts VLM architectures to Panoramic X-rays (OPGs). It is fine-tuned to detect caries, identify teeth (FDI notation), and classify dental implant brands. The model excels at fine-grained edge detection, tasks often missed by models trained on natural images.

SurgVLM [122] is a video-language model built upon the Qwen2.5-VL [123] backbone. It is trained on SurgVLM-DB, a massive database of 1.81 million frames and 7.79 million conversations. This dataset covers 16 surgical types and 18 anatomical structures. It is designed for temporal analysis and provides a strong baseline for surgical-video understanding. Some other models such as OphthaReason [124] is built on the MM-Retinal [124] dataset and utilizes reinforcement learning to perform step-by-step diagnostic reasoning for complex ophthalmic cases, moving beyond simple classification to explainable diagnosis.

E. MEDICAL AGENTIC VLMs

A growing line of work moves beyond single-pass inference toward agentic systems, in which a VLM functions as an orchestrator that dynamically selects and invokes specialized sub-modules rather than producing outputs through a fixed computational graph.

MedVersa [106] exemplifies this paradigm. Its Large Language Model orchestrator decides at inference time whether a query can be answered directly or requires delegating to a specialized vision module, such as a region-proposal network or a segmentation head. This dynamic routing allows MedVersa to handle heterogeneous tasks, including simultaneous report generation and anatomical landmark detection, within a single model invocation. Reported gains over static-pipeline baselines are most pronounced on tasks requiring multimodal reasoning rather than pattern matching.

Reasoning-to-segmentation pipelines such as LISA [130] demonstrate another agentic pattern: the model first generates a natural-language reasoning trace explaining *what* and *where* a pathological finding is, then uses that trace to condition a segmentation module (e.g., SAM [97] or MedSAM [129]) to produce a pixel-level mask. By decoupling high-level diagnostic reasoning from low-level localization, these systems reduce hallucinations in grounded outputs and achieve segmentation precision that purely generative models cannot match.

Tool-augmented diagnostic assistants extend this further by equipping VLMs with access to external knowledge sources, such as clinical guideline databases, drug interaction checkers, and PACS query APIs, at inference time. While most such systems remain at the prototype stage, they represent the direction toward clinically integrated AI, where the model operates as a component within a broader diagnostic workflow rather than as a standalone predictor.

F. TEMPORAL MULTIMODAL REASONING

Clinical diagnosis frequently requires integrating information across time: a follow-up chest CT must be interpreted in light of a prior scan, and a pathology report gains meaning when read alongside the patient's treatment history. Standard VLMs trained on single image-report pairs are not designed for this temporal reasoning task, and benchmark coverage of longitudinal multimodal scenarios remains sparse.

Med-MIM [89] is one of the few datasets specifically designed to evaluate multi-image visual reasoning, covering temporal understanding, cross-image comparison, co-reference resolution, and sequential reasoning. Models evaluated on Med-MIM must integrate information from multiple images presented jointly, which better approximates the follow-up imaging context encountered in clinical practice. MedVersa [106] likewise includes multi-image input modes in its evaluation protocol, demonstrating that agentic architectures with dynamic routing are better positioned than fixed single-image pipelines to handle longitudinal inputs.

Despite these early efforts, the field lacks a standardized longitudinal benchmark that pairs follow-up imaging with structured prior reports and lab results, evaluates interval-change detection explicitly, and measures clinical accuracy rather than just text-similarity proxies. Constructing such a benchmark is an important open problem for the community.

As Table 3 illustrates, three patterns stand out. First, *BLEU saturation* on MIMIC-CXR report generation: BLEU-1 scores cluster narrowly in the 22–33 range across very different architectures, suggesting that surface-level *n*-gram overlap is approaching a ceiling on this benchmark and is no longer discriminative between models. Second, *the dual-encoder vs. generative gap on VQA*: LLM-backed generative systems (Med-PaLM M, BiomedGPT) dominate SLAKE accuracy while contrastive dual-encoder models (BiomedCLIP, MedCLIP) remain competitive on retrieval but lag on open-ended VQA, reflecting the inherent advantage of generative decoding for free-text answer formats. Third, *benchmark fragmentation across modalities*: pathology models cluster on Quilt-1M and TCGA, radiology models on MIMIC-CXR and SLAKE, and 3D models on M3D, with very little cross-modality overlap that would permit direct architectural comparison. The prevalence of N/R entries itself is a finding: it reflects the absence of a community-standard evaluation suite spanning all five task domains, a gap discussed further in Section VII.

A comprehensive summary is provided in Table 4.

VII. CHALLENGES AND LIMITATIONS

VLMs have shown promise in integrating visual and textual data for medical applications, but their deployment in clinical practice faces significant challenges and limitations.

TABLE 3. Task-wise quantitative comparison of reviewed medical VLMs across five task domains. N/R = model not evaluated on this benchmark. All values are as reported in the original papers; direct cross-study comparison is limited by differing evaluation splits and prompt formats (see section IV).

Panel III-A: Radiology Report Generation

Model	BLEU-1	BLEU-4	ROUGE-L	METEOR	CIDEr	BERTScore	RadGraph-F1	Test Set
Med-PaLM M [64]	32.31	11.50	27.49	N/R	26.17	N/R	26.71	MIMIC-CXR
RadFM [109]	12.81†	N/R	18.22†	N/R	N/R	58.64‡	N/R	RadBench
CheXagent [65]	N/R	N/R	40.3§	N/R	N/R	N/R	14.6§	CheXpert / MIMIC-CXR
Med3DVLM [91]	36.88†	N/R	40.25†	36.42	N/R	88.11	N/R	M3D-Cap
MedVersa [106]	N/R	17.8¶	N/R	N/R	N/R	49.7¶	28.0¶	MIMIC-CXR

† Aggregate BLEU / ROUGE (not the *n*-gram-specific or -L variant). ‡ Reported as BERT-Sim, which differs from BERTScore. § ROUGE-L is Findings Summarization (MIMIC-CXR); RadGraph is Findings Generation (CheXpert) — two distinct tasks. ¶ Findings section.

Panel III-B: Visual Question Answering

Model	SLAKE Acc (%)	VQA-RAD Acc (%)	VQA-Med Acc (%)	Test Set
InstructBLIP [125]	55.8	42.8	22.2	SLAKE / VQA-RAD / VQA-Med
CogVLM [126]	83.2	56.8	62.1	SLAKE / VQA-RAD / VQA-Med
LLaVA-Med 1.5 [127]	55.9	52.9	39.1	SLAKE / VQA-RAD / VQA-Med
M3D [108]	54.4	49.7	26.3	SLAKE / VQA-RAD / VQA-Med
VividMed [95]	87.3	56.8	63.7	SLAKE / VQA-RAD / VQA-Med

† Accuracy is evaluated using an LLM judge (Llama-3-70B) as reported in [95]; all rows are taken from a single source under this identical protocol. RadFM is excluded from VQA-RAD / VQA-Med per the source, due to training-data overlap with these benchmarks.

Panel III-C: Image-Text Retrieval

Model	Recall@1	Recall@5	Recall@10	Mean Recall	Test Set
CONCH [113]	40.0	77.6	88.8	68.8	Source A (n=797)
PLIP [115]	5.6	19.8	30.7	18.7	Source A (n=797)
BiomedCLIP [101]	16.1	40.5	55.3	37.3	Source A (n=797)
OpenAICLIP [1]	1.8	4.6	8.4	4.9	Source A (n=797)

† Zero-shot text-to-image retrieval, Recall@K (%), reported on the CONCH "Source A" evaluation set (n=797) [113]. All four models evaluated under the identical protocol from that source.

Panel III-D: Zero-Shot Pathology Image Classification

Model	LC-Lung	LC-Colon	WSSSLUAD	SICAPv2	BMT	Average
PLIP [115]	88.79	78.38	86.21	50.61	33.33	67.47
CONCH [113]	88.93	97.35	65.14	47.36	40.67	67.89
PathGen-CLIP-L [128]	91.85	98.30	82.50	53.39	47.21	74.65
Patho-CLIP-L [66]	93.78	97.48	95.89	55.09	38.44	76.14

† Zero-shot pathology image classification accuracy (%) across five datasets, as reported in [66]. All models evaluated under the identical zero-shot protocol from that source. Average is the mean across the five datasets.

Panel III-E: Spine Segmentation (Intervertebral Disc & Vertebral Body)

2*Model	Intervertebral Disc			Vertebral Body		
	Dice	Sens.	Spec.	Dice	Sens.	Spec.
SAM (point) [97]	0.64	0.56	0.995	0.83	0.77	0.988
SAM (box) [97]	0.83	0.78	0.995	0.86	0.80	0.989
MedSAM (box) [129]	0.79	0.75	0.993	0.88	0.85	0.986

† Mean Dice, sensitivity, and specificity for segmenting intervertebral discs and vertebral bodies (standard deviations omitted for clarity; see source). RHD (robust Hausdorff distance) is also reported in the source but omitted here as it is a distance metric on a different scale. All models evaluated under the identical protocol from the source.

TABLE 4. Taxonomy of medical vision-language models: Mapping technical architectures to clinical utility.

Model Name	Architecture Paradigm	Adaptation Strategy	Clinical Task	Open Source	Highlights	Ref.
Imaging Modality: Multi-Modal / Foundation Models						
VividMed	Hybrid	3 Stage Training	Report Generation, VQA & Grounded Captioning	YES	Built upon CogVLM 17B; supports 2D and 3D imaging; uses parameter-efficient LoRA adaptation. Achieves 4.1% VQA accuracy increase on SLAKE and 8.5% improvement in macro CheXpert F1 for report generation.	[95]
Med-PaLM M	Large Multimodal generative model	Finetuning	VQA, Report Generation & Instruction-Following	NO	Built upon the PaLM-E architecture, scaling up to 562B parameters; supports clinical language, diverse biomedical imaging (radiology, pathology, dermatology, mammography, chest X-ray), and genomics; fine-tuned on MultiMedBench. Achieves over 14 percentage point increase in BLEU-1 on Slake-VQA and over 9 percentage point increase in Micro-F1-14 for chest X-ray report generation compared to prior state-of-the-art.	[64]
BiomedGPT	BERT style Encoder and GPT style Decoder	Pre-trained using large datasets, supports fine-tuning and Zero-shot prediction	VQA, Report Generation & Image-Text Retrieval	YES	Foundation model (33M-182M parameters) that processes diverse biomedical imaging modalities and clinical text. Pre-trained on 592K images and 183M text sentences across 25 tasks, it achieves SOTA performance on 16 benchmarks, including 97% accuracy on radiology datasets and outperforming GPT-4V in zero-shot visual question answering (54.7% vs 53.0%).	[100]
BiomedCLIP	Contrastive / Dual-Encoder	Pre-Training (PMC-15M)	Image-Text Retrieval & VQA	YES	OA MM foundation model (86M parameters) combining a Vision Transformer and PubMedBERT, pre-trained on PMC-15M. It achieves SOTA performance with 77% top-5 retrieval accuracy and 62% on VQA-RAD, outperforming general CLIP and radiology-specific models while enabling privacy-preserving analysis of patient data.	[101]
Imaging Modality: Radiology (X-Ray, CT, MRI, Ultrasound)						
Med3DVLM	Hybrid combines contrastive pretraining with generative modeling	3 stage training	Image-text retrieval, radiology report generation, and VQA on 3D CT volumes.	YES	Achieves SOTA performance across retrieval, report generation, and VQA tasks on the M3D dataset while maintaining computational efficiency.	[91]
RadFM	Generative	2 stage training	Report Generation & VQA	YES	14B-parameter generalist radiology foundation model that unifies 2D and 3D medical image understanding with multi-image, interleaved text input, trained on 16M scans across diverse modalities and 5000+ diseases. It outperforms existing open-source medical VLMs and GPT-4V on human evaluation across VQA, report generation, and rationale diagnosis tasks.	[109]
FetalCLIP	Dual Encoders for images and texts	Pre-trained, supports fine-tuning and Inference	Image-Text Retrieval & Grounded Captioning	NOT YET	CLIP-style vision–language foundation model trained on over 210k fetal ultrasound image–text pairs to learn universal fetal representations. It achieves SOTA on view classification, gestational age estimation, CHD detection, and fetal structure segmentation, with clinically interpretable internal representations.	[110]
Imaging Modality: Pathology (Histopathology & Digestive)						
CONCH	Hybrid; Contrastive and Generative	Multi-stage pre-training	Image-Text Retrieval, Grounded Captioning & VQA	YES	It is a hybrid contrastive-generative visual-language foundation model for histopathology, pretrained on 1.17M image–caption pairs with domain-specific unimodal initialization, achieving state-of-the-art across 14 benchmarks spanning zero-shot/few-shot classification, retrieval, segmentation, and captioning.	[113]
PathALIGN	Hybrid	Pre-Training & Fine-Tuning	Image-Text Retrieval & Report Generation	NO	Aligns whole slide histopathology images with diagnostic text from pathology reports using 350,000+ WSIs, enabling cross-modal retrieval, automatic report generation (78% accuracy rated by pathologists), and WSI classification through a BLIP-2 framework with frozen patch encoders and LLMs.	[117]
Imaging Modality: Visible Light / Surface (Derm, Ophtho, Dental)						
FLAIR	Contrastive	Pre-Training & Multiple adaptation options	Image-Text Retrieval & VQA	YES	Foundation VLM for retinal fundus images that integrates expert clinical knowledge by mapping 101 categorical disease labels to descriptive text prompts compiled from medical literature, achieving superior zero-shot generalization and few-shot adaptation compared to generalist models like CLIP and BiomedCLIP.	[118]
Chakrabarti et al.	Hybrid	2 stage training (Knowledge distillation)	Report Generation	NO	A two-stage knowledge distillation framework for automated skin cancer diagnostic report generation. The approach achieves 78% accuracy rated by pathologists.	[131]

A. DATA SCARCITY AND DOMAIN ADAPTATION

A major obstacle for medical VLMs is the persistent scarcity of large, diverse, and high-quality annotated datasets. Most publicly available datasets are small, static, and often lack the breadth needed to capture the full spectrum of diseases, imaging modalities, and patient demographics encountered in real-world clinical practice [94], [132]. This limitation is especially acute for rare or newly emerging diseases, where there may be few or no labeled examples available for model training or adaptation, severely restricting the ability of VLMs to generalize or respond to novel clinical scenarios [67].

The domain gap between general-purpose VLMs typically trained on natural images and generic text and specialized medical tasks further worsen this issue, as these models often lack the nuanced understanding required for accurate interpretation of complex medical data [94], [133]. Annotation of medical data is also expensive and time-consuming, requiring expert input, which further limits the creation of large, labeled datasets [132]. Recent efforts to address these challenges include transfer learning, disease-informed adaptation, and the use of synthetic data generation (such as domain-adapted generative models) to augment training sets, but these approaches only partially mitigate the underlying data scarcity [91]. Additionally, federated learning and collaborative data-sharing frameworks are being explored to increase dataset diversity while preserving patient privacy [79]. Among the reviewed studies, this challenge is clearly visible: FetalCLIP [110] trained a fetal ultrasound foundation model on only 210,000 image–text pairs, an order of magnitude smaller than natural-image corpora, and still achieved competitive zero-shot performance by combining contrastive pre-training with domain-specific curation; while MedCLIP [38] addressed the false-negative problem endemic to small clinical datasets through unpaired contrastive learning on separate image and text streams. Despite these advances, data scarcity and domain adaptation remain fundamental barriers to the widespread clinical adoption and reliability of medical VLMs.

B. GENERALIZATION, ROBUSTNESS, AND REASONING

Although medical VLMs have achieved impressive results in benchmark datasets, their ability to generalize and remain robust in real-world clinical environments is still limited [7]. Models often perform well on curated, clean datasets but experience significant performance degradation when exposed to noisy, corrupted, or out-of-distribution data, such as images with artifacts or from different imaging devices [134]. Robustness to domain shifts, imaging artifacts, and diverse clinical settings is not yet guaranteed, raising concerns about their reliability in practice. Furthermore, VLMs frequently struggle with complex clinical reasoning, such as integrating multimodal information, performing quantitative or spatial reasoning, and generating clinically coherent answers [74]. While reinforcement learning, structured prompting, and diverse training data have shown promise in improving reasoning and generalization, these methods are not yet sufficient to close the gap between model and clinician-level performance [135]. The lack of standardized evaluation metrics and comprehensive robustness benchmarks further complicates the assessment of model reliability. As a result, there is a pressing need for more rigorous, fine-grained evaluation protocols and robust adaptation strategies to ensure safe and effective deployment of VLMs in clinical settings [136].

A particularly acute manifestation of this generalization gap is the near-absence of prospective clinical validation among the reviewed studies. Of the 35 studies included in this review, only 3 included any form of prospective or external validation in a live clinical setting, with the remainder relying exclusively on retrospective held-out splits of existing datasets. This is a critical concern for regulatory readiness: the FDA's Software as a Medical Device (SaMD) pathway and the EU AI Act's high-risk provisions both require evidence of real-world generalization that retrospective benchmarks cannot provide [137]. Studies such as CheXagent [7], [65] demonstrate strong benchmark performance but report no prospective arm, illustrating the gap between technical maturity and clinical deployability.

C. BIAS, FAIRNESS, AND ETHICAL CONCERNS

Bias and fairness remain central challenges for medical VLMs, with significant implications for equitable healthcare delivery. Numerous studies have demonstrated that VLMs can perpetuate or even amplify demographic biases present in their training data, leading to disparities in diagnostic accuracy and clinical recommendations across groups defined by race, gender, ethnicity, language, age, and socioeconomic status [138], [139]. For example, expert-level VLMs have been shown to underdiagnose marginalized groups, with intersectional subgroups such as Black female patients experiencing the highest rates of underdiagnosis [140].

These biases are not limited to a single attribute but can span multiple demographic dimensions, and may be further exacerbated by the encoding of demographic information within model representations. The sources of bias are multifaceted, including imbalanced datasets, algorithmic design choices, and broader structural inequities in healthcare systems [141]. Biases can manifest in both model outputs and the underlying probability distributions, sometimes resulting in miscalibrated confidence or unfair treatment recommendations [142]. The impact of these biases is far-reaching, potentially intensifying existing health disparities and eroding trust in AI-driven medical tools.

To address these issues, researchers have developed a range of mitigation strategies, such as fairness-oriented model architectures, prompt engineering, post-hoc debiasing methods, and the use of fairness-aware loss functions [143]. New datasets with detailed demographic annotations, like Harvard-FairVLMed, are enabling more granular fairness

evaluations and the development of targeted interventions [76]. Additionally, frameworks such as FairMed and FairCLIP have demonstrated measurable improvements in fairness metrics across demographic groups, while multimodal chain-of-thought prompting and participatory evaluation approaches offer promising directions for both open and closed-source models [144]. Despite these advances, challenges remain. There is a lack of standardized benchmarks and reporting practices for bias and fairness, and many mitigation strategies are still in early stages or have only been tested in limited settings [145]. Ethical concerns extend beyond technical bias, encompassing transparency, accountability, privacy, and the need for ongoing monitoring and community engagement throughout the model lifecycle [141]. As VLMs become more integrated into clinical workflows, it is essential to ensure that fairness and equity are prioritized at every stage from data collection and model development to deployment and post-market surveillance.

Drawing on the evidence base across the reviewed studies, we identify eight best practices for fairness-aware medical VLM development: (1) *demographic-stratified evaluation*report performance separately for each demographic subgroup rather than aggregate only [139]; (2) *fairness-aware loss functions* incorporate group-regularization terms (e.g., DPO variants, equalized-odds constraints) during fine-tuning [76]; (3) *diverse pre-training corpora* supplement standard corpora with under-represented-population data (e.g., Fitzpatrick-17k for skin tone diversity) to reduce representation gaps before adaptation; (4) *intersectional auditing* evaluate not only single-attribute subgroups but joint subgroups (e.g., race $\times$ sex) where disparities can be substantially larger [140]; (5) *post-hoc calibration* apply temperature scaling or Platt scaling per subgroup to correct miscalibrated confidence; (6) *participatory dataset design* involve clinicians and patient communities from under-represented groups in dataset curation to surface label biases before training; (7) *standardized fairness reporting* adopt a common reporting schema (metrics, subgroup definitions, confidence intervals) to enable cross-study comparisons, following the toolbox proposed by [144]; and (8) *post-market fairness monitoring* treat fairness as an ongoing operational requirement, with periodic re-evaluation as deployment populations shift.

D. EXPLAINABILITY

Explainability is critical for the trustworthy adoption of VLMs in clinical practice. The ''closed box'' nature of many VLMs limits clinician trust and hinders regulatory approval. Within the reviewed literature and the broader field, three distinct tiers of explainability can be distinguished.

The first tier comprises *attribution-based explanations* post-hoc saliency maps, GradCAM overlays, and SHAP (SHapley Additive exPlanations) values that identify which image regions or text tokens most influenced a prediction [146], [147]. While these methods are widely deployed, they answer only the question of *where* the model attended, not *why* it reached its conclusion, limiting their usefulness to clinicians who need to verify diagnostic reasoning rather than pixel salience.

The second tier consists of *reasoning-trace explanations* chain-of-thought outputs, step-by-step diagnostic narratives, and structured rationales generated alongside the prediction. These provide a richer window into model reasoning but introduce their own risks: traces can be fluent yet factually incorrect, a form of plausible-sounding hallucination that may be harder to detect than a wrong label.

The third and most clinically actionable tier is *localization-anchored explanation*, which integrates lesion localization, a reasoning trace, and a structured report into a single coherent output. As argued in recent work on AI in breast imaging [148], clinicians require not merely a heatmap but a complete chain connecting model predictions to specific lesion locations and reasoning processes. Most reviewed VLMs do not yet achieve this integrated form. Among those that approach it, LISA-style reasoning-to-segmentation pipelines [130] ground diagnostic language in pixel-level segmentation masks; RetinaVLM couples ophthalmic report generation with lesion bounding-box outputs; and PathAlign aligns pathology report sentences with their corresponding tissue-patch evidence. These systems represent the emerging frontier of clinically verifiable explanation and serve as concrete targets for the field.

Data-privacy considerations that complement explainability principally Federated Learning and associated privacy-preserving training techniques are treated alongside dataset-bias mitigation in Section V; the combined goal of interpretable and privacy-preserving deployment frames the trustworthy-AI agenda for medical VLMs [79], [149], [150].

E. COMMERCIAL LANDSCAPE AND REGULATORY HURDLES

VLMs in healthcare is accelerating, with major technology companies and startups integrating large language models into clinical products and digital health services. However, this rapid adoption brings unique regulatory and business challenges. Regulatory ambiguity persists regarding the approval and oversight of VLM-based medical devices, especially as existing frameworks struggle to keep pace with the evolving capabilities of generative AI. Key regulatory concerns include patient data privacy, intellectual property rights, medical malpractice liability, quality control, standardization, and informed consent. For example, the integration of VLMs into already-approved medical devices raises questions about how regulatory bodies (e.g. FDA) will evaluate and monitor these hybrid systems. Furthermore, the lack of clear guidelines for transparency, risk management, and non-discrimination complicates the deployment of VLMs in high-risk clinical settings. The commercial landscape is also shaped by the need for robust audit procedures and the challenge of ensuring that AI-driven models do not perpetuate healthcare disparities or compromise patient safety. As a

result, collaboration between regulators, technology developers, and healthcare providers is essential to establish effective oversight and foster responsible innovation in the commercial deployment of VLMs in medicine [137], [151], [152].

A critical but under-discussed regulatory distinction concerns the *type* of VLM architecture seeking approval. Purely closed box generative models which produce outputs through opaque neural transformations with no auditable intermediate steps face the steepest regulatory scrutiny, as neither the FDA's SaMD framework nor the EU AI Act provides a straightforward pathway for systems that cannot articulate the basis for a high-risk clinical decision. By contrast, agentic and modular VLM architectures, in which discrete reasoning steps, tool calls, and verification loops are logged and inspectable, are better positioned to satisfy the transparency and traceability requirements that regulators are converging on. Systems such as MedVersa and LISA-style reasoning-to-segmentation pipelines [130] demonstrate that structuring the model's decision process as an auditable sequence of steps can simultaneously improve clinical reliability and reduce the regulatory surface area of uncertainty. This architectural distinction is likely to become an increasingly important factor in the commercialization trajectory of medical VLMs.

F. CLINICAL WORKFLOW INTEGRATION: PACS, RIS, AND EHR SYSTEMS

Beyond regulatory approval, the practical integration of VLMs into existing hospital infrastructure represents a distinct and substantial barrier. Most clinical imaging workflows are organized around PACS and RIS, which manage image storage, retrieval, worklist scheduling, and report routing. These systems operate under strict interoperability standards (DICOM, HL7 FHIR) and institutional data governance policies that were not designed with AI inference in mind. Inserting a VLM into this pipeline requires addressing latency constraints (radiologists expect sub-second image loading), output formatting (structured radiology reports must conform to department templates), authentication and audit-logging requirements, and failsafe behavior when the model is unavailable or produces a low-confidence output.

Among the reviewed studies, very few explicitly report integration with live clinical systems. MedVersa demonstrates a step toward workflow awareness by accepting heterogeneous clinical inputs images, structured notes, and task descriptors within a single inference call, reducing the need for bespoke pre-processing pipelines at the point of integration. RetinaVLM similarly pairs report generation with lesion localization outputs in a format compatible with ophthalmology reporting standards. However, neither study reports end-to-end deployment within a production PACS or RIS environment, and the gap between research prototype and clinical-grade integration tool remains large. Future work should target this gap directly, including evaluation of VLM outputs within existing radiologist review queues, measurement of time-to-report with and without AI assistance, and assessment of model behavior under real-world data-quality conditions such as missing sequences, corrupted DICOM headers, or incomplete clinical histories.

G. FROM MODEL-CENTRIC TO SYSTEM-CENTRIC EVALUATION

The dominant evaluation paradigm for medical VLMs remains model-centric: performance is measured on static held-out splits of existing datasets, and progress is declared when benchmark numbers improve. While this approach is tractable and reproducible, it systematically under-measures the properties that matter most for clinical adoption namely, whether the model improves care-pathway outcomes when embedded in a live clinical system.

A more informative evaluation framework would adopt a system-centric perspective, moving from static benchmark metrics to a closed-loop assessment that follows the full operational cycle: sample collection, automated report generation, clinician review, and feedback-based calibration. This framing is analogous to the design philosophy demonstrated in automated biosensing platforms [153], where validation is performed end-to-end within the operational loop rather than on isolated test sets. Applied to medical VLMs, a system-centric benchmark suite would need to include: (i) *workflow integration metrics* latency, throughput, and failover behavior within PACS/RIS pipelines; (ii) *time-to-decision* whether AI-assisted review shortens or lengthens the overall diagnostic timeline; (iii) *downstream care-pathway effects* whether model outputs correlate with changes in treatment decisions or clinical outcomes; and (iv) *feedback-loop fidelity* whether the system can incorporate clinician corrections to improve over time without catastrophic forgetting.

Of the 35 studies reviewed here, none report system-level evaluation across all four dimensions. This is not a criticism of individual authors constructing such evaluations requires institutional partnerships and longitudinal study designs that lie beyond the scope of a single research group but it does identify a clear structural gap in the field that funding agencies, hospital research offices, and benchmark consortia are best positioned to address.

H. HALLUCINATION DETECTION AND MITIGATION

Hallucination the generation of fluent but factually incorrect clinical content is one of the most consequential failure modes of generative VLMs in medicine. Unlike classification errors, hallucinated outputs can be syntactically well-formed, internally consistent, and difficult for a non-expert to identify, making them particularly dangerous in automated reporting pipelines where clinician oversight may be intermittent. Several technical approaches to detecting and mitigating hallucinations have emerged in the reviewed literature and the broader field. RAG reduces hallucination by grounding generation in retrieved evidence from trusted corpora such as radiology report databases or clinical guidelines, making it harder for the model to fabricate findings that have no

TABLE 5. Barrier coverage matrix: Which deployment barriers each reviewed model explicitly addresses (✓), Partially addresses (∼), or does not address (–). DS = data scarcity; EV = external Validation; Do = domain shift; Ha = hallucination; Ex = Explainability; Fa = fairness; WI = Workflow Integration. Grading is based on each model's abstract and stated contributions, applying a strict bar in which ✓ denotes a dedicated method, experiment, or contribution targeting that barrier; ∼ denotes an explicit but non-dedicated mention; and – denotes no mention.

Model	DS	EV	Do	Ha	Ex	Fa	WI
MedCLIP [38]	✓	–	∼	–	–	–	–
BiomedGPT [100]	∼	–	✓	–	–	∼	∼
FetalCLIP [110]	✓	–	✓	–	∼	–	–
CheXagent [65]	∼	–	–	–	∼	–	✓
FairCLIP [76]	–	–	–	–	–	✓	–
Med3DVLM [91]	–	–	∼	–	–	–	–
Med-PaLM M [64]	∼	–	✓	–	∼	–	–
RadFM [109]	∼	–	✓	–	∼	–	–
MedVersa [106]	∼	✓	✓	–	∼	–	✓
CONCH [113]	✓	–	∼	–	–	–	∼
PathAlign [117]	∼	–	–	–	∼	–	✓
Patho-R1 [66]	–	–	∼	✓	✓	–	–
PA-LLaVA [114]	∼	–	∼	–	–	–	–
RetinaVLM [119]	–	–	–	–	∼	–	∼
SkinVL [120]	∼	–	∼	–	–	–	–
VividMed [95]	✓	–	∼	∼	✓	–	–
BiomedCLIP [101]	∼	–	✓	–	–	∼	–
XrayGPT [154]	∼	–	–	–	∼	–	–
SurgVLM [122]	∼	–	∼	–	∼	–	–

evidential basis [74]; retrieval-grounded radiology generation has been reported to reduce hallucination rates by roughly 18-25% relative to non-retrieval baselines, although the magnitude varies with the richness of the available structured context. Reasoning-to-segmentation pipelines impose an automatically checkable consistency constraint between spatial claims in text and pixel-level masks, so that any finding asserted in language must correspond to localized image evidence (see Section VI). Self-verification and cross-referencing loops, as implemented in agentic frameworks, allow a model to re-examine its own outputs against image evidence before committing to a final report. Post-hoc detection methods, including natural-language-inference-based fact-checking against structured medical knowledge graphs and consistency probing, are also being explored. As the barrier coverage matrix (Table 5) makes clear, however, explicit hallucination mitigation remains the exception rather than the rule among the reviewed models. Only Patho-R1 [66] treats hallucination control as a dedicated training objective, using reinforcement-learning reward shaping that rewards plausible step-by-step reasoning and penalizes hallucinated diagnostic steps. VividMed [95] addresses the problem only indirectly: by coupling report generation with segmentation masks and instance-level bounding boxes, it ties generated language to localized image evidence, which constrains ungrounded claims as a by-product of grounded generation rather than through an explicit hallucination objective. The remaining reviewed models including report-generation systems such as CheXagent [65] and generalist models such as Med-PaLM M [64] and MedVersa [106] report output-quality metrics but do not implement or measure a dedicated hallucination-control mechanism. This sparse coverage points to a deeper methodological gap. The grounding and verification based mechanisms above are best understood as ways of *reducing* the opportunity for fabrication, not as direct *measurements* of it. Conventional generation metrics (BLEU, ROUGE, BERTScore) and retrieval or grounding scores (Recall@K, gIoU) quantify text overlap or localization quality, not factual faithfulness, and are therefore poor proxies for clinical hallucination. The field still lacks a community-standard benchmark that explicitly measures whether each generated clinical statement is faithfully supported by the input image and the patient record. Establishing such a benchmark together with per-statement faithfulness scoring is an important open problem if hallucination is to be assessed rather than merely argued about.

Two influential frameworks from the general-domain literature point toward capabilities that medical VLMs increasingly require. LISA [130] introduces *reasoning segmentation*, generating a segmentation mask from a complex, implicit query by coupling the language-generation ability of a multimodal LLM with an embedding as mask mechanism, while FLAIR [118] learns fine-grained, text-conditioned local image representations from long, detailed captions, improving localized retrieval and zero-shot segmentation. Although both are trained on natural images and text rather than clinical data, their core ideas grounding language in pixel-level masks and aligning text to localized image regions map directly onto the clinical

needs for verifiable, region anchored outputs, suggesting strong potential for adaptation to medical applications such as grounded report generation and reasoning driven lesion delineation.

VIII. EMERGING PARADIGMS

The following subsections survey four frontier directions that have gained significant momentum in the 2024 to 2026 period, followed by a structured roadmap for researchers entering this field.

A. REASONING-CAPABLE VLMs

Standard medical VLMs generate answers in a single forward pass, with no explicit intermediate reasoning. Reasoning-capable VLMs extend this by producing structured chains of diagnostic inference identifying findings, applying clinical rules, and weighing differential diagnoses before committing to a final output. This paradigm, adapted from chain-of-thought (CoT) prompting in text-only LLMs, is particularly valuable in medical contexts where the *process* of reasoning is as important as the conclusion.

Two recent examples illustrate the trajectory. Patho-R1 [66] applies reinforcement-learning-based reward shaping to train a pathology-specialized VLM to generate step-by-step tissue-level reasoning chains, showing measurable gains in diagnostic coherence on patch-level classification tasks. OphthaReason [124], trained on the MM-Retinal-Reason dataset, demonstrates multi-step reasoning over retinal images combined with clinical history, bridging the gap between image-level findings and diagnosis-level conclusions.

Beyond generating reasoning traces, a further development is *self-reflective verification*: after producing an initial answer, the model re-evaluates its output against the image evidence, checking internal consistency before finalizing the response. While promising, this loop is computationally expensive and adds latency that may be unacceptable in high-throughput screening settings.

A critical open problem is the absence of appropriate evaluation metrics. Standard automatic metrics (BLEU, ROUGE, CIDEr) measure surface-level token overlap against a reference answer; they do not assess whether the reasoning steps are clinically valid, logically coherent, or faithfully grounded in the image. Emerging alternatives include process-reward scoring (assessing each reasoning step independently), step-faithfulness metrics (checking whether each stated finding is visually verifiable in the input), and structured benchmark suites such as MedReasonBench. Within the four-axis taxonomy of Figure 7, reasoning-capable VLMs primarily advance the *Architecture Paradigm* and *Clinical Task* axes: they shift the paradigm from standard encoder-decoder generation toward iterative, verifier-augmented generation, and they expand the task coverage to include multi-step differential reasoning rather than single-step classification or captioning.

B. LIGHTWEIGHT AND EDGE-DEPLOYABLE VLMs

The dominant VLMs discussed earlier in this review operate at 70B+ parameter scales on multi-GPU infrastructure that is inaccessible to the majority of clinical sites globally. A growing body of work instead targets models in the 1B-7B parameter range that can run on a single consumer GPU or, in some cases, an edge device while retaining meaningful medical capability. Within the reviewed literature, BiomedGPT [100] is a concrete example: its smaller checkpoints (down to 182M parameters) achieve competitive performance on several biomedical benchmarks at a fraction of the compute cost of generalist systems such as Med-PaLM M [64], demonstrating that a compact VLM can remain clinically useful. Beyond the medical domain, general-purpose efficiency efforts including open-weight small language models such as BioMistral [155] and modular clinical SLM strategies such as MediPhi [156] illustrate complementary routes to reducing model footprint, although these are predominantly text-only systems and would require a vision encoder and multimodal fine-tuning before they could perform image-based clinical tasks. A central enabling technique is parameter-efficient adaptation. Quantized Low-Rank Adaptation (QLoRA) applies 4-bit quantization to the backbone while fine-tuning only low-rank adapters, and is widely reported to reduce memory requirements by roughly 60-70% relative to full half-precision loading while preserving most of the task-specific performance gained during fine-tuning [52]. This places 7B-class medical VLMs within reach of a single consumer GPU and substantially lowers the hardware barrier for academic groups and resource-limited clinical sites. Parameter count and memory footprint, however, are only part of the deployment picture. In mobile screening units and remote or low-resource primary-care clinics, a model must function under intermittent network connectivity, on consumer-grade hardware with limited cooling and power budgets, and with highly variable input quality from portable imaging devices. Drawing on the design principle of ‘‘controllability, mobility, and precise loading’’ demonstrated in mobile biosensing platforms [157], we argue that the next generation of clinical VLMs should be evaluated not only on benchmark accuracy but also on deployment robustness: graceful degradation under input corruption, low-latency inference on constrained hardware, and reliable behavior under intermittent connectivity. A practical obstacle to this kind of evaluation is the absence of any standardized, cross-hardware reporting protocol for medical VLMs. Existing papers report inference cost when they report it at all under heterogeneous and often unstated conditions (different hardware, batch sizes, sequence lengths, quantization schemes, and whether or not input prefill is included), which makes published latency and throughput figures effectively non-comparable across studies. We therefore identify the establishment of a common reporting standard specifying a fixed workload, a defined set of representative hardware tiers (e.g., a data-center GPU, a consumer GPU, and an edge accelerator), and consistent treatment of prefill, precision, and memory accounting as an

important and currently unmet need for the community. Until such a standard exists, claims of ''edge readiness'' should be treated with caution and verified under the specific hardware and input conditions of the intended deployment site.

C. VISUAL CHAINING AND GROUNDED REASONING

A key insight from recent literature is that while generalist VLMs excel at clinical reasoning, they often lack the spatial granularity required for precise localization. To bridge this gap, a growing trend toward *visual chaining* decouples high-level reasoning from low-level execution. In these composite frameworks, the VLM functions as a semantic orchestrator rather than a standalone classifier, delegating spatial tasks to specialized sub-modules such as the Segment Anything Model (SAM) [97] or MedSAM [129] to achieve geometric accuracy that end-to-end models struggle to match. The canonical reasoning-to-segmentation example (LISA) is discussed in Section VI, Medical Agentic VLMs; the broader point here is that visual chaining as a *paradigm* not any single instantiation is what unlocks the integration of natural-language diagnostics with the pixel-level precision needed for pre-surgical planning and grounded reporting.

D. OPEN-SOURCE VLMs AND AGENTIC CLINICAL WORKFLOWS

While initial breakthroughs were driven by massive, proprietary models, the deployment landscape reveals a significant pivot toward Small Language Models (SLMs) as a direct counter-measure to the privacy and regulatory bottlenecks inherent in cloud-based APIs. Open-weights models in the 7B–13B parameter range such as BioMistral [155] and MediPhi [156] can effectively rival larger proprietary counterparts when fine-tuned on high-quality biomedical corpora. The development of open-source VLMs such as Molmo [6] further advances transparency and reproducibility: making model weights and training code publicly available addresses the opacity of proprietary API-based systems and enables genuine external auditing of model behavior. Efficiency techniques such as Quantized Low-Rank Adaptation (QLoRA) lower the hardware barrier for academic research groups, allowing detailed examination of structural and training pipeline choices that is not possible with API-based systems.

The next frontier in clinical deployment lies in the transition from static prediction to dynamic agentic workflows. Traditional models operate on a ''System 1'' basis providing immediate, unverified outputs which poses significant risks in high-stakes screening scenarios. Emerging agentic frameworks introduce a ''System 2'' reflexive process, where the AI iteratively plans, executes, and critiques its own diagnostic steps. Systems employing this self-corrective mechanism such as cross-referencing findings with prior scans or refining segmentation masks through iterative feedback demonstrate improved reliability. This shift toward auditable self-verification offers a practical pathway toward trustworthy clinical decision support.

E. ROADMAP FOR RESEARCHERS ENTERING THE FIELD

The following seven research directions represent high-priority open problems accessible to researchers at early-career or group level. For each direction we provide: (a) the core open problem, (b) the prerequisite skills and data resources needed to begin, and (c) a concrete entry-point benchmark.

1) **Reasoning-VLM Evaluation.**
   (a) BLEU/ROUGE metrics do not assess step-wise reasoning correctness or clinical coherence of diagnostic chains produced by CoT-style VLMs.
   (b) Familiarity with LLM evaluation frameworks (EleutherAI LM Harness); access to OphthaReason dataset [124] or PathMMU.
   (c) MM-Retinal-Reason with process-reward scoring on intermediate reasoning steps.
2) **Federated Medical VLM Training.**
   (a) Communication-efficient federated learning for heterogeneous medical imaging modalities across institutions without sharing raw DICOM data.
   (b) FL framework experience (Flower, FedML); simulated multi-site split of MIMIC-CXR and CheXpert [79].
   (c) Simulated federated MIMIC-CXR report generation with per-site performance gap as a primary metric.
3) **Fairness-Aware Adaptation.**
   (a) Standard PEFT methods do not incorporate group-fairness constraints; post-hoc debiasing can disrupt vision-language alignment.
   (b) Harvard-FairVLMed dataset [76]; fairness metrics (equalized odds, AUC gap across demographic subgroups).
   (c) FairVLMed ophthalmic classification fairness evaluation suite with AUC gap between demographic subgroups as the optimization target.
4) **Agentic Clinical Workflows.**
   (a) No standardized benchmark exists for multi-step clinical agent tasks spanning history retrieval, image analysis, guideline lookup, and structured report generation within a single pipeline.
   (b) LLM tool-use API experience; EHR simulation environment; MIMIC-III/IV for patient history data.
   (c) MultiMedEval composite benchmark spanning report generation and clinical VQA components.
5) **Edge-Deployable Medical SLMs.**
   (a) Quantization of vision encoders (not only LLM decoders) degrades localization accuracy on high-resolution pathology slides, with no standardized cross-hardware reporting protocol.
   (b) QLoRA / bitsandbytes toolchain; access to edge hardware (NVIDIA Jetson or consumer GPU); BioMistral [155] or BiomedGPT [100] weights.
   (c) VQA-RAD or SLAKE accuracy at 4-bit QLoRA on a consumer GPU, compared against a full-precision baseline.

6) **Prospective Clinical Validation.**
   (a) No standardized protocol exists for embedding VLM outputs in a radiologist review workflow and measuring time-to-decision or downstream patient-outcome effects prospectively.
   (b) IRB approval; hospital or clinic partnership; familiarity with PACS APIs (OHIF, Orthanc) and HL7 FHIR for data capture.
   (c) A 50-100 case pilot integrating CheXagent [65] or equivalent into a test PACS environment, with time-to-report as the primary outcome.
7) **Integrated Localization-Reasoning-Reporting Systems.**
   (a) No single model currently achieves all three tiers of clinically actionable explainability (attribution, reasoning trace, and localization-anchored reporting) within a unified architecture.
   (b) LISA/SAM codebase familiarity [130]; paired segmentation-report datasets (MS-CXR-T, RGRG); multi-task training experience.
   (c) MS-CXR grounded report generation (simultaneous localization and language generation) or RGRG region-guided radiology report generation.

F. LIMITATIONS OF THIS REVIEW

This review has several explicit limitations that bound the strength of its conclusions. First, the included set of 35 studies skews toward English-language publications and toward radiology and pathology, reflecting the demographic and modality distribution of the underlying literature rather than a complete clinical landscape; ultrasound, dental, and surgical-video VLMs are underrepresented. Second, our quality-rubric scoring relied on the metrics reported by the original authors and could not adjudicate methodological inconsistencies (e.g., differing evaluation splits, varying prompt formats) that prevent strict apples-to-apples comparison across studies; the cross-study numbers in Table 3 should therefore be read as indicative rather than head-to-head. Third, the barrier coverage assessment in Table 5 is based on each model's stated contributions and reported capabilities rather than on independent re-implementation, and the distinction between fully addressing (✓) and partially addressing (∼) a barrier necessarily involves reviewer judgment. Fourth, the absence of a standardized cross-hardware and cross-model reporting protocol meant that inference-cost figures across studies were not directly comparable; we therefore did not attempt to tabulate quantitative deployment-cost measurements, and instead treat standardized efficiency reporting as an open community need (Section VIII-B). Fifth, the rapidly moving nature of the field means that models released after the January 2026 search cutoff are not included. Finally, dual-reviewer screening was applied to a 10% audit sample rather than the full set, which is a standard trade-off but does mean that single-reviewer error in the unaudited portion cannot be fully excluded. We flag these limitations not as obstacles to the review's conclusions but as the boundary conditions within which those conclusions should be interpreted.

IX. CONCLUSION

In this survey, we systematically analyze the advancement of VLMs in medical applications. VLMs show strong potential in medical AI, offering improved diagnostic accuracy, faster interpretation, and broader access to expert-level image analysis especially in settings where specialists are scarce. The field is shifting from monolithic classifiers toward modular architectures where VLMs act as reasoning engines, delegating tasks like segmentation to specialized sub-modules. This design directly addresses clinical needs for both diagnostic reasoning and spatial precision. At the same time, smaller open-weights models are gaining traction, enabling local deployment that meets privacy requirements while maintaining strong performance. The emergence of agentic workflows where systems iteratively verify and refine their outputs offers a practical mechanism to reduce false positives in high-stakes screening. However, key challenges remain. Limited dataset scale and diversity restrict generalization, particularly for rare conditions and underrepresented populations. Prospective validation showing real-world patient benefit is still lacking, and current regulatory frameworks do not fit the dynamic nature of these systems. Without careful attention to fairness and performance monitoring across patient groups, these tools risk increasing rather than reducing healthcare disparities. Clinical adoption will likely be gradual, with VLMs augmenting rather than replacing physician judgment. Moving these systems toward trusted clinical use will require ongoing collaboration between technologists, clinicians, regulators, and patients grounded in rigorous validation and a clear focus on safety, effectiveness, and equity.

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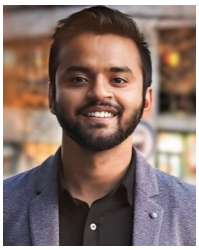
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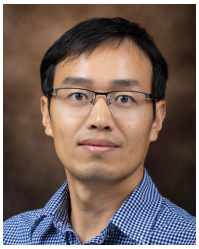
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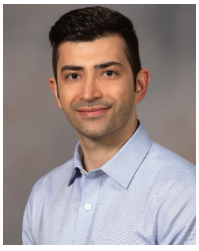
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